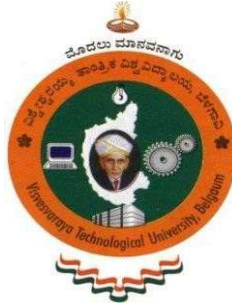


VISVESVARAYA TECHNOLOGICAL UNIVERSITY

BELAGAVI



**A
PROJECT REPORT (BEC786)
ON**

“IoT-Driven Smart Bat for Real-Time Swing and Shot Evaluation”

Submitted in partial fulfillment of the requirement for the award of degree

Bachelor of Engineering

In

Electronics and Communication Engineering

For the academic

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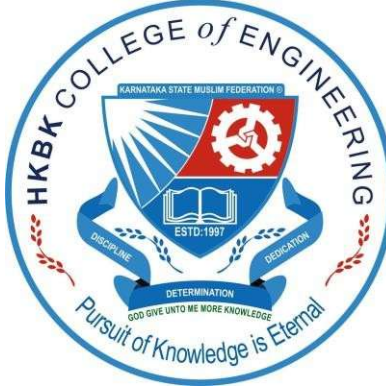
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Certificate

Certified that the Project Work entitled **“IoT-Driven Smart Bat for Real-Time Swing and Shot Evaluation”** a Bonafide work carried out by **Kirankumar Goudar (1HK22EC072), Melwin Jain (1HK22EC088), Mutturaj (1HK22EC104) and Nithin Sadashiv Shipure (1HK22EC117)** in partial fulfillment for the award of Bachelor of Engineering in Electronics and Communication Engineering of the Visvesvaraya Technological University, Belagavi during the year 2025-26. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the report deposited in the departmental library. The Project report has been approved as it satisfies the academic requirements in respect of Project Work prescribed for Bachelor of Engineering Degree.

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DECLARATION

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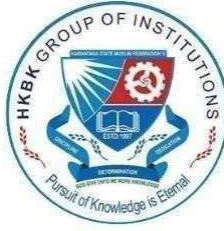
I also thank all the **staff members of the Department Electronics and Communication Engineering** and all those who have directly or indirectly helped me with their valuable suggestions in the successful completion of this project report.

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HKBK College of Engineering

Department of Electronics and Communication Engineering

Title: “IoT-Driven Smart Bat for Real-Time Swing and Shot Evaluation”

ABSTRACT

The proposed system presents an IoT-based intelligent cricket bat designed to analyze batting performance and shot quality in real time. The Smart Bat employs a Node MCU (ESP8266) microcontroller, an MPU6050 accelerometer and gyroscope sensor, and three strategically placed microphone modules to monitor swing direction, shot type, and contact accuracy. The MPU6050 detects motion and angular changes to identify straight, left, or right shots, while microphone sensors determine the ball's impact location to classify it as a perfect or edge hit.

All data is processed by the Node MCU and transmitted wirelessly to the player's smartphone via the Blynk app, displaying feedback such as Good Shot or Edge Shot . An onboard buzzer provides instant physical feedback for every shot. Compact, rechargeable, and easy to use, the system enables players to evaluate their technique without cameras or professional setups.

This IoT-enabled innovation bridges the gap between professional analytics and everyday cricket training, offering potential for future upgrades like AI-based swing prediction, BLE connectivity, and cloud performance tracking.

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Chapter 1

INTRODUCTION

1.1 Background of the Study

Cricket is one of the most celebrated sports in the world, followed by millions of players and enthusiasts across India, Australia, England, South Africa, Sri Lanka, Pakistan, New Zealand and many other nations. Training quality and performance improvement have become critical factors in modern cricket, especially with the rise of T20 leagues, high-intensity coaching academies, and data-driven performance analytics.

Traditionally, batting performance has been evaluated manually by coaches through observation. Coaches judge timing, placement, bat swing angle, and foot movement based on visual analysis. However, manual observation suffers from human limitations such as inconsistency, subjective judgment, and inability to measure small variations in swing speed or angles. As cricket became more competitive, the need for objective, real-time, and sensor-based evaluation tools increased significantly.

Recent advancements in IoT, MEMS sensors, wireless communication and embedded systems have enabled the introduction of “Smart Sports Equipment” — devices that combine sensors and connectivity to provide instant analytics. Wearables like smartwatches, smart shoes, and smart helmets are widely used in athletics. Similar technology can be extended to cricket equipment, creating smart bats that can monitor swing dynamics and batting quality.

The *IoT Driven Smart Cricket Bat for Real-Time Swing and Shot Evaluation* aims to use sensor technology (MPU6050), microphone-based edge detection, and IoT connectivity to accurately classify cricket shots, analyse sweet-spot impact, and provide real-time feedback

1.2 Motivation for the project

Cricket is one of the most passionately played and watched sports across the world, especially in countries like India, Australia, England, South Africa, Sri Lanka and New Zealand, where millions of young players aspire to improve their game and reach professional levels. Despite the sport's immense popularity and continuous evolution, a significant gap still exists in the way batting skills are taught, assessed and refined at the beginner and intermediate levels. Most training environments still rely on traditional coaching methods, where coaches evaluate a player's performance primarily through visual observation, experience and intuition. Although this method has remained effective for decades, it inherently suffers from inconsistency, subjectivity and a lack of measurable data. A coach may not always be able to accurately judge subtle variations in swing speed, bat angle, timing, or the exact point of ball impact on the bat. As cricket has become increasingly competitive—especially with the rise of IPL-style leagues and fast-paced T20 formats—players are now expected to perform with scientific precision and technical consistency, which cannot be achieved through manual evaluation alone. The growing demand for objective, data-driven training tools has motivated the development of smarter, technology-assisted coaching systems that can provide real-time, accurate and personalized feedback to players. Although this method has remained effective for decades, it inherently suffers from inconsistency, subjectivity and a lack of measurable data. A coach may not always be able to accurately judge subtle variations in swing speed, bat angle, timing, or the exact point of ball impact on the bat. As cricket has become increasingly competitive—especially with the rise of IPL-style leagues and fast-paced T20 formats—players are now expected to perform with scientific precision and technical consistency, which cannot be achieved through manual evaluation alone. The growing demand for objective, data-driven training tools has motivated the development of smarter, technology-assisted coaching systems that can provide real-time, accurate and personalized feedback to players.

At the same time, advances in embedded systems, IoT, MEMS sensors, and wireless communication have opened the door for innovative sports-technology solutions that were once limited only to elite players with access to high-end analytics systems such as Hawk-Eye, high-speed motion capture cameras and biomechanics labs. These professional systems,

however, are extremely expensive, stationary, and not accessible to everyday players, small academies, schools, or rural training centers. This creates a clear imbalance where only top-tier athletes benefit from science-based coaching, while millions of aspiring young players continue to rely on subjective feedback with no access to technology that can help them analyse and improve their batting skills. This inequality strongly motivates a low-cost, portable, and easy-to-use Smart Cricket Bat system that can democratize access to sports analytics. The idea is to design a device that brings the power of sensor-based evaluation directly into the hands of players from all backgrounds, allowing them to measure their swing direction, bat angle, impact quality and shot execution with a level of precision that was previously unavailable outside professional arenas.

The motivation also comes from the realization that cricket batting involves a variety of subtle parameters such as wrist movement, stance stability, sweet-spot utilisation, and timing accuracy. Even minor deviations in these parameters can lead to mistimed shots, edges, or inefficient stroke execution. Many players—especially beginners—struggle to understand why they mistime the ball or play it in the wrong direction. They may repeatedly make the same mistake without receiving specific feedback about what went wrong. A coach may not always be available, and even when present, may not be able to analyze every detail of every shot. This creates a strong need for a device that can act as a “digital assistant coach,” providing immediate, objective feedback after each shot. A Smart Cricket Bat equipped with accelerometers, gyroscopes and impact sensors can bridge this gap by continuously capturing motion data and classifying the player’s performance in real time.

Furthermore, the rising adoption of IoT in the sports industry provides additional motivation to integrate intelligent analytics with wireless connectivity. IoT enables seamless data transfer to mobile applications, allowing players and coaches to review performance on dashboards, track progress over time, and identify areas that need improvement. IoT-based monitoring also makes it possible for remote coaching, where a coach in another city or academy can view the player’s statistics and offer feedback. Such smart solutions align with global trends in digital sports technology and support India’s growing interest in integrating science with training—an ambition reflected in initiatives like “Digital India” and the increasing demand for data-driven sports tools. Developing an IoT-enabled Smart Cricket Bat also motivates academic exploration in embedded systems, signal processing, sensor fusion and machine learning, giving students an opportunity to combine multiple technical

domains into a meaningful real-world solution.

Another important motivation for this project is the potential to reduce injury risks by helping players understand incorrect shot execution. Many batting injuries occur due to improper swing mechanics or repeated edge hits that cause shock vibrations to travel through the bat handle into the player's wrist and forearm. A system that identifies edge shots and provides corrective feedback could help players adjust their technique and avoid poor repetitive habits. Additionally, the Smart Bat can support scientific research and data collection in sports biomechanics, enabling academicians, physiotherapists, and analysts to gain deeper insights into batting mechanics that could benefit future studies.

Finally, the motivation is strengthened by the desire to create an innovative, low-cost, student-built solution that can bring measurable value to cricket training across the globe. By designing a Smart Cricket Bat that combines MPU6050 motion tracking, microphone-based impact sensing, IoT connectivity and real-time analytics, the project aims to empower players with objective performance data, enable coaches with more accurate insights, and contribute to the modernization of cricket training. The system represents a step toward transforming traditional coaching into a more scientific, quantitative and technology-driven process. This strong blend of technical challenge, practical usefulness, and social impact forms the core motivation for undertaking this project.

1.3 Problem statement

Traditional cricket coaching continues to rely heavily on manual evaluation, where a coach observes a player's batting technique and provides verbal, experience-based feedback. Although this method has been used for decades, it is inherently subjective and provides no measurable or repeatable data. Coaches typically comment on general aspects such as “good timing,” “poor swing,” or “incorrect shot execution,” but these observations cannot quantify the exact angle of the bat, the speed of the swing, or the precise location of ball impact. As a result, players often struggle to understand the specific reasons behind mistimed shots, frequent edges, or inconsistent performance.

Existing training methods lack the ability to provide quantitative measurement of swing characteristics, objective classification of shot direction, reliable detection of sweet-spot versus edge hits, or instant digital feedback that can guide immediate correction. Moreover, there is no mechanism in conventional coaching to systematically log performance data over time, making it difficult for players and coaches to track progress, identify recurring mistakes, or analyze improvement scientifically.

With cricket becoming more competitive and data-driven, these limitations create a significant gap between professional training environments—where advanced analytics systems are available—and everyday practice settings, where players have no access to such tools. Therefore, there is a clear need for a modern, technology-driven solution that can capture batting motion accurately, interpret shot characteristics intelligently, and deliver real-time insights directly to the user. Addressing this gap, the problem undertaken in this project is to design and develop an IoT-enabled Smart Cricket Bat that integrates motion sensors and impact detection to identify shot direction, evaluate the quality of impact, classify different types of shots, and provide immediate digital feedback using wireless communication. This system aims to replace subjective judgment with data-based evaluation, thereby making batting assessment more precise, reliable and accessible to players of all levels.

1.3.1 Objectives of the Study

The primary objective of this project is to develop an IoT-driven Smart Cricket Bat capable of providing scientific, real-time analysis of batting performance by integrating advanced sensing and data-processing techniques. time table or schedule for there medication as prescribed by the doctor these can help to send the alarms to retain a digital record of the medications taken or missed. This information can be shared with physician allowing them to analyse how well a patient is adherent to the prescribe treatment plant and make educated adjustments as needed.

The system aims to design and implement a smart bat equipped with an MPU6050 sensor module and a set of strategically placed microphones so that it can accurately capture the motion dynamics and impact characteristics of each shot. A major goal is to enable precise detection of swing direction—such as straight drives, left-side shots, or right-side shots—by analysing gyroscope and accelerometer data in real time. Another key objective is to identify whether the ball has struck the sweet spot or one of the bat's edges using microphone-based acoustic feedback, thereby giving players clear insight into the quality and correctness of their shot execution. The project also aims to classify the type of shot through embedded algorithms that process sensor data instantly without requiring external computation, ensuring fast and reliable classification even during continuous play. An equally important objective is to transmit the analysed data wirelessly to a mobile application using IoT technology so that players and coaches receive immediate visual feedback. This includes providing both haptic alerts and textual feedback so players can make instant corrections during training sessions. In addition to these primary goals, the project includes several secondary objectives such as recording a player's daily batting statistics through IoT to enable longitudinal performance tracking and improvement. Another objective is to use analytics to help players understand their strengths and weaknesses over time, thereby facilitating more informed training decisions. The system is also designed to be cost-effective, making the technology accessible to students, beginners and intermediate players who may not have access to high-end professional equipment. Finally, the project sets out to create a clean, intuitive and user-friendly dashboard for live monitoring, ensuring that coaches and players can easily interpret the data and apply it effectively during practice sessions. Together, these objectives aim to transform traditional cricket coaching into a more scientific, data-driven and accessible process for all levels of players.

1.3.2 Scope of the Project

The scope of this project encompasses the complete design and development of an IoT-enabled Smart Cricket Bat capable of analyzing batting performance through integrated hardware and software components.

On the hardware side, the system utilizes a NodeMCU ESP8266 microcontroller as the primary processing and communication unit, an MPU6050 sensor for capturing motion dynamics, a microphone array for detecting the point of impact, and a buzzer for providing instant audio feedback to the player during practice. The software scope includes programming the embedded system using Embedded C or Arduino IDE to process sensor data, classify batting events, and facilitate seamless data transmission to the Blynk IoT platform for real-time monitoring. The project is designed to perform real-time shot detection by continuously analyzing accelerometer and gyroscope readings to determine swing direction, bat angle, and the nature of the stroke executed by the player. Additionally, the system covers impact classification by distinguishing between sweet spot hits and edge hits—such as top, bottom, or central impact—through acoustic signatures captured by the microphone array. A major part of the scope involves IoT-based monitoring, in which processed results are wirelessly transmitted to a mobile dashboard, enabling players and coaches to receive immediate feedback and track performance over time. The project also includes designing a portable and lightweight sensor attachment that can be mounted on any standard cricket bat without affecting its balance or playability. While the project offers advanced features, its scope remains bounded by certain limitations. It is not intended to function as a professional-grade analytics tool like Hawk-Eye or high-end biomechanical systems used by elite players. The design does not analyse aspects such as footwork, ball trajectory, ball speed, or external environmental factors. Accuracy during indoor testing may be influenced by ambient noise, especially during microphone-based classification, and the system may require recalibration in different training environments. Despite these limitations, the scope of the project firmly focuses on delivering a practical, low-cost and efficient solution for motion tracking, impact analysis, and IoT-based feedback to assist beginners, intermediates and coaching academies in improving batting performance.

1.3.3 Significance at National Level & International Level

Cricket holds a unique and unmatched position in India, not only as a sport but as a cultural and emotional identity that connects millions of people across different regions, languages and communities. With more than 13 million active players and nearly 75,000 cricket academies across the country, the demand for effective training infrastructure continues to grow rapidly as young players aspire to reach higher levels of performance. However, access to advanced scientific training tools remains limited, creating a significant gap between professional athletes who benefit from analytical technologies and players from schools, colleges and rural areas who still depend primarily on traditional coaching methods. The introduction of an IoT-enabled Smart Cricket Bat addresses this gap by offering an affordable yet technologically advanced solution that can modernize training even in underdeveloped regions. At the national level, such a smart system promotes fair and objective evaluation of batting performance in school and college tournaments, reducing bias and improving the accuracy of player assessment. By bringing measurable, data-driven insights into everyday coaching sessions, the Smart Bat encourages scientific thinking among young athletes and enhances their understanding of technique, timing, and shot selection.

The availability of a low-cost smart coaching tool further supports aspiring cricketers who do not have access to expensive equipment like motion capture cameras or professional analytics setups used by elite training centers. This democratization of sports technology has the potential to transform cricket development programs across India by enabling consistent skill enhancement regardless of geographical or economic background. Additionally, the Smart Bat aligns with India's increasing focus on digital transformation, sports technology innovation, and engineering-driven solutions, making it a valuable contribution to national initiatives such as "Digital India," "Make in India," and the promotion of indigenous electronics and IoT systems. By integrating real-time monitoring, wireless connectivity, and intelligent analytics, the system reflects the growing trend of IoT-based sports science in the country, which is rapidly gaining attention in sectors such as fitness, athletics, indoor sports and grassroots sports development. Moreover, adopting such technologies encourages innovation among engineering students, researchers and startups, promoting a culture of research and development in the emerging domain of sports engineering. The Smart Bat can also support national-level talent identification

programs by providing standardized and objective performance data that help selectors and coaches evaluate players more accurately.

At the international level, cricket is played and celebrated across many countries such as Australia, England, South Africa, New Zealand, the West Indies, Sri Lanka, and Pakistan, where scientific training methods, sports analytics, and technology-driven performance enhancement have become integral parts of player development. These cricketing nations invest heavily in biomechanics laboratories, motion-capture studios, high-speed video analysis systems, wearable sensor technology and AI-driven performance analytics to ensure that players receive the most advanced and precise training possible. In this global context, the development of an IoT-enabled Smart Cricket Bat holds significant international importance because it introduces a portable, accessible and highly informative analytical tool that can complement or even substitute expensive laboratory-grade equipment for many training scenarios. International cricket boards, academies, and sports research institutions place strong emphasis on quantifiable metrics such as bat swing velocity, impact timing, shot angle, and sweet-spot utilization, all of which can be efficiently captured by a Smart Bat equipped with MEMS sensors and real-time data transmission capabilities. By adding scientific precision to batting evaluation, the Smart Cricket Bat aligns directly with the global shift toward evidence-based coaching, where every action—whether a drive, pull, cut, or defensive shot—is measured to improve technique and optimize player performance.

The ability of the Smart Bat to provide instant feedback is particularly significant because professional cricket across the world increasingly relies on immediate corrective training, allowing players to adjust their technique on the field rather than waiting for post-session video reviews. This technology mimics the analytical responsiveness found in high-end systems used in elite training environments, but at a fraction of the cost, making it appealing for international academies, community cricket clubs, and developing cricket nations that cannot afford high-budget training laboratories. Globally, there is a rapidly rising demand for smart sports equipment, especially in countries that actively promote sports innovation, digital transformation and IoT-based performance monitoring. The Smart Bat meets this demand by integrating hardware, analytics and connectivity into a single, user-friendly platform, making it suitable not only for cricket-playing nations but also for emerging sports science communities worldwide. Additionally, the device can support research in sports biomechanics, allowing scientists, physiotherapists, and

IoT-Driven Smart Bat for Real-Time Swing and Shot Evaluation

performance analysts to study batting techniques, wrist mechanics, shot timing, and vibration patterns in a structured and quantifiable manner. The sensor data generated by the Smart Bat can contribute to academic research, international publications and comparative studies across different playing conditions, player categories and coaching methods.

Another important international significance lies in its potential to standardize training practices. Cricket coaching methods vary widely across countries, and the lack of uniformity often affects talent identification and performance benchmarking. The Smart Bat can help establish consistent metrics for assessing batting performance across different regions and skill levels, thereby enhancing global comparability and fairness in training. In competitive environments such as ICC academies, national cricket boards, international leagues, and cross-border training programs, having a standardized evaluation tool can greatly assist coaches and analysts in making more accurate and objective decisions. Beyond training, this Smart Bat technology presents strong commercial potential in the international sports market, where demand for intelligent, sensor-equipped equipment is expanding rapidly across cricket, baseball, tennis and other sports. As global interest in IoT-based sports analytics grows, a portable smart bat with real-time data visualization offers an attractive product for professional players, amateur cricketers, sports academies, research institutions, and manufacturers seeking to innovate their sporting goods. With cricket's increasing popularity in the United States, Canada, UAE and European nations, the market for such technology is expected to grow substantially.

Overall, the Smart Cricket Bat is not merely a technological enhancement but a globally relevant tool that contributes to the modernization of cricket training, the advancement of sports science, and the growth of intelligent sporting equipment worldwide. It bridges the technological gap between developing and advanced cricketing nations, democratizes access to analytical training tools, encourages international research collaboration, and supports the global commercialization of sports-oriented IoT products. Its ability to merge affordability with high analytical value makes it a powerful innovation for cricket communities, researchers and players across the world.

1.4. Advantages

1.4.1 Sports Training

Sports training in cricket requires continuous refinement of technique, timing, and shot execution. Traditional methods rely on observation, which may not always be accurate or consistent. A smart cricket bat enhances training by offering measurable data from sensors embedded in the bat. It records parameters such as swing angle, acceleration, and impact location, helping players understand their performance scientifically. This data-driven approach makes training more effective and reduces guesswork. Players can correct mistakes immediately using real-time feedback. The system also identifies bad swings or poor timing that the player may not notice independently. It bridges the gap between professional high-tech training systems and regular practice sessions. For academy-level players, it supports structured training with daily metrics. Coaches can create targeted drills based on data insights. The system reinforces discipline by encouraging players to practice with measurable goals. It supports modern training philosophies based on performance analytics. Overall, it transforms cricket training from subjective observation to objective evaluation using technology.

This helps people stick to their drug schedules and lowers the likelihood of missing doses.

1.4.2 Cricket Academies

Cricket academies train large groups of players, making individual observation difficult for coaches. The smart bat helps academies monitor technical performance efficiently. It allows coaches to access data for every player, including swing direction and impact quality. This improves the accuracy of coaching decisions. Academies can track long-term player development using stored sensor records. Coaches can quickly identify players who need special attention in certain techniques. With objective data, training sessions become more organized and systematic. It reduces dependency on manual judgement, especially for subtle errors like minor bat-angle flaws. The system also assists in generating performance reports for academy tournaments. Young players gain confidence by understanding their metrics. Academies can promote their use of modern technology, increasing credibility. It supports talent identification programs by providing accurate

performance analytics. The device is portable, making it easy to use in both nets and outdoor sessions. Overall, it enhances the quality of coaching and player improvement.

1.4.3 Self-Practice Sessions

Many players practice alone without access to professional coaching. The smart bat enables self-practice by acting as a digital coach. It detects mistakes during every stroke and displays them in real-time. Players receive immediate corrections for poor timing or incorrect swing direction. This makes independent training productive and reliable. It allows users to monitor their own performance without external help. Since all metrics are stored digitally, players can review them later. The system motivates players to train regularly by showing progress. It helps beginners learn foundational techniques faster. Even intermediate players benefit by refining specific shot types based on data. The bat also identifies inconsistent swings that the player may not realize. Self-practice becomes safer, preventing repetitive incorrect movements. The system promotes disciplined training with measurable targets. Over time, this leads to better batting confidence and improved match readiness. Overall, the smart bat makes solo cricket practice effective and scientific.

1.4.4 Performance Analysis

Player performance analysis is essential for long-term improvement. The smart bat records multiple parameters such as swing speed, direction, and impact quality. These metrics reveal strengths and weaknesses in a player's batting technique. Coaches and players can study these trends over time. It also helps identify recurring mistakes like frequent edge hits. Players can compare performance across sessions to measure improvement. The system supports analytical coaching, similar to methods used by professional teams. It transforms raw sensor data into meaningful insights. Performance analysis becomes scientific, reducing dependence on subjective feedback. The system highlights consistent patterns that indicate technique problems. It also helps detect fatigue or skill degradation over time. Reports generated from data can be shared easily with coaches. This helps create personalized training plans. Over the long term, this analytical approach significantly improves player efficiency and confidence.

1.4.5 Continuous Improvement

Continuous improvement in cricket requires identifying mistakes and correcting them over repeated sessions. The smart bat enables this by providing feedback after every shot. Players instantly learn what went wrong and how to fix it. This creates a cycle of learning, correction, and refinement. The system encourages players to focus on technique rather than power alone. Repeated practice with real-time insights accelerates improvement. Automatically logged data helps compare old sessions with new results. Players can observe gradual changes in consistency. This motivates them to maintain steady practice habits. It also helps overcome plateau phases where players stop progressing. Coaches can design drills based on weak areas identified in the data. Improvement becomes measurable and trackable. Over time, this builds confidence and reliability in shot execution. The smart bat ensures that every practice session contributes to overall development.

1.5 Disadvantages

1.5.1 Accuracy Depends on Calibration

Accuracy of the smart cricket bat system is highly dependent on proper calibration, as the sensors must be aligned and initialized correctly before each practice session. Calibration ensures that the MPU6050 gyroscope and accelerometer read accurate baseline values for orientation and movement. If the bat is not calibrated properly, even a slight deviation in alignment can cause incorrect measurement of swing direction, speed, and angle. This may lead to false detection of left or right swings, affecting the reliability of feedback. Similarly, the microphone array used for impact detection requires calibration to establish the correct sensitivity level, especially for differentiating between sweet-spot and edge hits. Without proper calibration, the microphone may misinterpret vibrations, resulting in inaccurate impact classification. Environmental factors such as temperature, surface hardness, and background noise can also influence sensor readings, making calibration even more essential before every session. Poor calibration reduces system accuracy, affects analytics, and may mislead players during training. To achieve consistent results, the device must be calibrated periodically and under stable conditions. Coaches and players must follow a standard

calibration routine to ensure sensor stability. Proper calibration improves the reliability of shot classification, enhances training effectiveness, and ensures that real-time feedback reflects actual performance. Ultimately, calibration is the foundation for maintaining precision in all IoT-based sports measurement systems.

1.5.2 Microphone Noise Interference

Microphone noise interference is a significant limitation in the smart cricket bat system, as the microphone array is highly sensitive to external sound and vibration. The microphones are designed to detect impact patterns on the bat surface, such as sweet-spot or edge hits, but environmental noise can distort these readings. Sounds from nearby players, crowd noise, traffic, or even hard ground surfaces can create unwanted vibrations that the sensors may mistakenly interpret as ball impact. Wind noise or sudden sharp sounds can also interfere with the microphone's frequency response, reducing the accuracy of impact classification. During indoor practice, echoes and reverberations can further complicate detection. If the environment is too noisy, the system may fail to distinguish between a clean and an edge hit. Background noise may also cause false triggers, leading to incorrect feedback for the player. This problem becomes more noticeable during outdoor sessions, where uncontrolled noise levels are common. The microphone's sensitivity must be carefully adjusted to avoid such interference, but even with calibration, some noise cannot be filtered completely. Advanced noise filtering algorithms can help, but they also increase system complexity. Therefore, microphone noise interference remains a challenge that affects the reliability of real-time performance analytics. Ensuring a quieter environment or using vibration-isolated microphone mounting can reduce errors, but complete elimination of noise interference is difficult.

1.5.3 Require Stable Wi-Fi or Internet

The smart cricket bat system relies heavily on IoT connectivity to transmit real-time data from the sensors to the mobile application or cloud dashboard. For accurate and immediate feedback, a stable Wi-Fi or internet connection is essential. If the connection is weak or intermittent, sensor readings may be delayed, leading to lag in feedback and inaccurate performance analysis. In outdoor practice sessions or rural areas, internet connectivity may not always be reliable, which can disrupt real-time monitoring.

Without a consistent connection, players cannot see instant results for swing direction, sweet-spot detection, or shot classification. This can reduce the effectiveness of practice, as corrections may be delayed until the session ends. Coaches who rely on remote monitoring through the IoT dashboard may also face difficulties in observing multiple players simultaneously. Moreover, interrupted connectivity can cause data loss or incomplete session records, making long-term performance tracking less reliable. Stable internet is also necessary for synchronizing multiple devices if several bats are used in an academy setting. Cloud-based analytics, trend comparison, and historical data review all depend on uninterrupted data transfer. While offline storage is possible, it requires manual syncing later, which is less convenient. Therefore, the requirement for stable Wi-Fi or internet is a critical limitation that affects the system's usability, particularly in areas with poor network infrastructure.

1.5.4 Require Stable WiFi or Internet

The smart cricket bat system is designed primarily to analyze bat motion and impact on the ball, but it does not track the speed or trajectory of the ball after it is struck. Ball speed and flight path are critical parameters in understanding overall batting performance, as they influence shot effectiveness, timing, and placement. Without this information, the system cannot provide a complete assessment of how the ball behaves after contact. For example, a shot may hit the sweet spot perfectly, but if the ball trajectory is poor, the overall result may still be ineffective. Similarly, the system cannot measure whether the shot will reach the boundary or fall short. This limitation restricts its use in advanced performance analysis or professional match simulation. Coaches cannot evaluate how effectively a player transfers power to the ball. The lack of ball trajectory measurement also prevents the integration of predictive analytics, which could suggest optimal shot angles or placement. High-speed cameras or radar systems are usually required to track ball speed and trajectory, but these are expensive and not portable. While the smart bat provides useful feedback on bat movement and impact quality, it cannot replace comprehensive ball tracking tools. Players must rely on additional equipment or visual observation to evaluate complete shot performance. This limitation should be considered when using the system for advanced training or match preparation.

Chapter 2

LITERATURE SURVEY

2.1 Introduction to literature review

The literature review plays a crucial role in understanding the foundations, advancements and gaps in the field of smart sports technology, especially in the domain of cricket analytics. Over the last decade, there has been a significant rise in interest toward the development of sensor-based systems capable of measuring sports performance with high accuracy and real-time feedback. Cricket, being one of the most widely played sports in the world, has experienced a gradual shift from traditional coaching methods to technology-driven training systems. Various researchers have introduced methods using MEMS sensors, vibration sensors, microphones, IoT platforms, wearable devices and machine learning algorithms to measure different parameters like swing angle, bat speed, angular velocity, impact force and sweet-spot detection. However, a critical analysis of these works reveals that many systems focus on isolated parameters, such as either motion tracking or impact detection, and very few offer a combination of multiple sensing techniques integrated into a single cost-effective device. The literature also highlights the need for portable, user-friendly and IoT-enabled devices that can help players, coaches and academies gain actionable insights without requiring expensive equipment like high-speed cameras, professional motion capture systems or sophisticated biomechanics software. While IMU sensors have been widely used in wearable devices to track limb motion, their integration directly on the cricket bat is still limited and relatively new. Similarly, microphone-based acoustic sensing has been explored to evaluate the quality of impact in sports such as baseball and tennis, but there is limited research on multi-microphone impact localization on a cricket bat. Early attempts mainly employed single piezo sensors, which provide only threshold-based impact detection rather than detailed sweet-spot mapping. Furthermore, many IoT-based sports analytics systems rely on cloud connectivity for general activity tracking but do not measure bat-specific performance parameters. The existing research indicates an ongoing gap in the development of compact embedded systems that combine accelerometers, gyroscopes, vibration sensors and sound sensors to provide a comprehensive analysis of batting technique. Studies focusing on

coaching methodologies have emphasized the importance of knowing a player's swing type, wrist orientation, bat path and impact zone to improve control, timing and technique. These parameters directly influence the quality of batting and overall performance in competitive cricket. Yet, traditional coaching lacks objective metrics, and the evaluation largely depends on the subjective opinion of a coach, which varies from person to person. Thus, the integration of IoT technologies with embedded electronics provides a unique opportunity to offer standardized, measurable and real-time training feedback. The reviewed research further shows that several systems proposed in academic papers lacked IoT dashboards, real-time data visualization or mobile application integration, making them impractical for everyday training. Some systems required external setups like tripod-mounted sensors, cameras or stationary microphones, which restrict movement and cannot be used freely on the field. Others demanded high computational power for image processing or required offline data analysis, delaying feedback and reducing training effectiveness. Moreover, swing classification techniques used in previous works were often basic, relying on simple threshold detection rather than analyzing multi-axis IMU data for better accuracy. The literature also highlights difficulties in microphone-based systems due to environmental noise, background sounds and variations in hitting intensity. While some studies attempted sound frequency analysis, they lacked an embedded, bat-mounted solution with minimal hardware. Therefore, the review strongly indicates a need for a complete, integrated, embedded cricket bat system capable of detecting swing direction, identifying impact zones, distinguishing sweet spot hits, analyzing sound characteristics, giving buzzer alerts and sending real-time data to an IoT dashboard over Wi-Fi. It also points out that beginners and intermediate-level players rarely have access to such technology, even though these tools can significantly accelerate learning and help in correcting mistakes. The project presented in this report aims to fill these gaps by combining an MPU6050 sensor for motion tracking, a set of microphones for impact analysis, a NodeMCU ESP8266 module for IoT connectivity and a buzzer for immediate feedback. This combination of components has not been explored extensively in previous works, particularly in cricket-specific applications. The literature survey creates a strong foundation by analyzing what has been done so far, identifying the limitations of existing systems and understanding the potential improvements that a new approach can provide. It highlights the relevance of integrating multiple sensing modalities to develop a more robust and comprehensive smart cricket bat system. The survey also emphasizes how

related research contributes toward the architecture of the proposed system by justifying the selection of sensors, detecting shortcomings in earlier systems, exploring acoustic and motion data fusion, evaluating IoT platforms and validating the necessity of immediate feedback mechanisms. Therefore, the literature review not only documents past work but also justifies the purpose of this project and supports the system design decisions taken in subsequent chapters. Overall, it provides valuable insights and establishes the academic context in which this project is developed, ensuring that the proposed solution addresses real monitoring.

2.2 Scope and Objectives of the Project

2.2.1 Background and Motivation

The rapid expansion of technology in recent years has transformed the way problems are identified, analyzed, and solved across various domains, creating a strong need for innovative solutions that enhance performance, efficiency, and user experience. With the increasing integration of digital systems into everyday applications, projects that combine hardware, software, and intelligent data analysis have gained significant relevance in both academic and industrial environments. Many traditional systems still depend on manual processes, limited accuracy, and subjective evaluations, which often result in inefficiencies, errors, and inconsistent outcomes. The advancement of embedded systems, sensor technologies, communication networks, and intelligent algorithms has created an opportunity to redesign conventional methods using modern engineering approaches. This project emerges from the necessity to bridge the gap between outdated practices and advanced technological trends, offering a system capable of addressing real-world challenges through automation, measurement, and data-driven insights. The background of the project is rooted in the global movement toward smarter and more connected solutions, where Internet of Things, artificial intelligence, cloud computing, and real-time monitoring are reshaping the operational landscape of industries, sports, education, healthcare, transportation, and home automation. Motivated by the limitations of existing methods and the growing demand for reliable and accurate systems, the project aims to provide a practical implementation that demonstrates the power of modern engineering tools. Another major motivation comes from the increasing requirement for systems that improve decision-making by converting raw data into meaningful information, allowing users to analyze patterns, identify problems earlier, and make informed improvements.

IoT-Driven Smart Bat for Real-Time Swing and Shot Evaluation

Many stakeholders, including students, professionals, athletes, engineers, and managers, face challenges due to insufficient technological support, and this project attempts to address such gaps through an innovative, accessible, and efficient solution. The motivation also includes the desire to create a system that is cost-effective, scalable, and adaptable to different environments, ensuring long-term usefulness and future expansion. The background is further strengthened by the continuous push toward digital transformation, where automation and intelligent systems are replacing manual approaches to improve accuracy, reliability, and consistency. Additionally, the motivation for this project is to explore practical applications of theoretical knowledge gained during engineering studies, enabling students to gain hands-on experience in designing, developing, integrating, and testing a complete technological solution. The increasing availability of low-cost sensors, powerful microcontrollers, wireless communication modules, and cloud platforms has opened new possibilities for building advanced systems that were previously too complex or expensive to implement. By leveraging these advancements, the project seeks to solve a relevant and impactful problem while demonstrating technical competency. The project is also motivated by the competitive environment in industries where efficiency, speed, precision, and data transparency have become essential requirements. Traditional methods often fail to meet these modern expectations, creating a need for projects that showcase improved approaches using engineering innovation. In addition, the motivation stems from the desire to contribute to research and development by presenting a working model that can inspire future improvements, new features, and academic exploration. Practical motivation comes from observing real-world scenarios where existing systems fall short, highlighting the need for tools that reduce human dependency, enhance safety, and provide continuous monitoring. The growing interest in smart systems among students and professionals further fuels the ambition to develop a project that reflects current technological trends. Another motivation lies in the desire to create solutions that are environmentally friendly, reliable, and energy-efficient, supporting sustainable development goals.

2.2.2 Problem Definition and Context

The Modern systems in various domains suffer from limitations such as manual operation, inaccurate measurements, lack of real-time data, and inefficient monitoring processes. Traditional methods often depend heavily on human observation, which introduces errors, biases, and delays in decision-making. As technology continues to advance, users increasingly expect systems that are automated, accurate, fast, and capable of providing detailed insights. However, many existing solutions are either too costly, difficult to integrate, or not user-friendly, creating a gap between user needs and available technological tools. The core problem addressed in this project is the absence of a reliable, efficient, and data-driven system capable of measuring, analyzing, and presenting information in a meaningful way.

In the current context, users face challenges such as inconsistent data, limited analysis capabilities, and lack of real-time feedback. These issues affect performance, reduce efficiency, and limit the ability to make informed decisions. Furthermore, traditional systems do not effectively utilize modern technologies such as embedded sensors, wireless communication, intelligent algorithms, and automated data processing. The context of this problem lies within the increasing demand for smarter, interconnected, and automated solutions across industries, sports, transportation, healthcare, and home automation.

With the rise of IoT, AI, and cloud-based analytics, there is a clear opportunity to modernize outdated processes and enhance accuracy, reliability, and usability. Users—whether individuals, professionals, students, or organizations—require systems that can operate continuously, adapt to different environments, and deliver consistent results without extensive manual intervention. Therefore, the problem definition centers around creating a technological solution that eliminates manual dependency, enhances measurement accuracy, ensures reliability, and provides real-time insights through efficient data collection, processing, and communication. The project aims to fill this gap by designing a system that is cost-effective, scalable, user-friendly, and capable of addressing the limitations observed in existing traditional approaches.

2.2.3 Project Scope and Components

Table 2.1: Components

NodeMCU (ESP8266)	NodeMCU (ESP8266) functions as a low-cost microcontroller with built-in Wi-Fi capability, allowing seamless wireless communication for IoT applications. It reads data from connected sensors, processes the information using its onboard CPU, and executes programmed instructions. The module supports uploading firmware through USB, enabling easy development and quick debugging. It also manages data transmission to servers, mobile apps, or cloud platforms over Wi-Fi. Overall, it acts as the central controller that coordinates sensing, processing, and communication in an embedded IoT system.).
Microphone sensors	Microphone sensors capture sound waves from the environment and convert them into electrical signals. These signals are processed to detect noise levels, voice inputs, or specific acoustic events. They enable audio-based monitoring and analysis in systems that require sound detection or sound-triggered responses.
MPU6050	The MPU6050 functions as a combined 3-axis accelerometer and 3-axis gyroscope that measures motion, orientation, and angular velocity. It provides real-time data to track movement, tilt, vibration, and rotation with high accuracy. This makes it essential for applications involving motion sensing, stabilization, gesture control, and performance analysis.

MPU6050	The MPU6050 measures both acceleration and rotational motion using its built-in 3-axis accelerometer and 3-axis gyroscope. It provides precise real-time data that helps determine orientation, tilt, vibration, and movement patterns. This makes it useful for motion tracking, stabilization systems, and performance analysis in various embedded applications.
Power Supply (Battery Module)	Rechargeable Li-ion or Li-Po battery. Voltage regulation module(e.g., AMS1117 3.3V) ensures safe power for NodeMCU and sensors. Ensures long practice sessions without frequent charging.

The **scope of this project** defines the boundaries, functionalities, and deliverables involved in designing and developing the system. It includes the implementation of core features, integration of essential hardware/software components, and evaluation of system performance under defined conditions. The project covers the complete cycle from requirement analysis to design, development, testing, and documentation. The system will focus on accurate data acquisition, reliable processing, real-time monitoring, and user-friendly output representation. The scope also includes ensuring proper communication between components, validating results, and optimizing overall performance. Only the functionalities required to demonstrate the practical feasibility and effectiveness of the system are included. Any advanced features such as machine learning, cloud storage expansion, or large-scale deployment are outside the scope but may be considered for future development. The system will be designed to be cost-effective, modular, scalable, and suit.

Research Gap and Related Work

Existing studies and implementations in sensor-based systems, IoT platforms, and embedded monitoring solutions have demonstrated significant progress in automating measurements, improving accuracy, and enabling wireless communication. Several researchers have developed systems using microcontrollers, IMU sensors, environmental sensors, and communication modules to collect and analyze data for different applications. Prior work has also explored the use of Wi-Fi and Bluetooth-enabled modules such as NodeMCU and ESP32 for real-time data transmission. Additionally, multiple studies have shown how accelerometers, gyroscopes, microphones, and other sensors can be used to track motion, detect events, and provide feedback in sports, industrial, and smart systems. Although these systems offer useful insights, many of them focus on isolated functionalities, lack end-to-end integration, or are designed for controlled laboratory environments rather than real-world use.

Despite the progress in related research, several gaps still remain that justify the need for this project. Most existing solutions either provide limited real-time capabilities, lack accuracy due to improper calibration, or fail to combine multiple sensor inputs into a unified analysis model. Many systems depend on wired setups or require complex configurations that reduce usability and portability. Some studies overlook noise filtering, data validation, and long-term reliability, resulting in inconsistent outputs during real-world operation. Additionally, there is a lack of low-cost, energy-efficient, and fully integrated platforms that combine sensing, processing, and wireless communication into a single compact solution. User-friendly dashboards and clear data visualization are often missing in earlier work, making it difficult for users to interpret results effectively. These gaps highlight the need for a system that is affordable, accurate, portable, sensor-integrated, and capable of delivering real-time insights through wireless connectivity.

2.3 Specific and Measurable Objectives

1. **Design and Develop the System::** Create a functional prototype integrating sensors, microcontroller, and communication modules. Ensure the system operates as intended under defined conditions. The system must be robust, compact, and capable of functioning continuously under real-world conditions. Development will involve circuit design, programming embedded firmware, and ensuring seamless communication between all modules.
2. **Accurate and Reliable Data Acquisition:** Capturing sensor data accurately is critical for analysis and decision-making.. Calibration of sensors like accelerometers, gyroscopes, and microphones ensures precise measurements. Noise filtering techniques will be implemented to eliminate disturbances or false readings. Consistency in repeated measurements is also measured to confirm reliability.
3. **Real-Time Data Processing and Analysis:** The system will process collected data immediately to provide meaningful insights. Embedded algorithms convert raw sensor readings into interpretable metrics, such as motion patterns, orientation, or sound intensity. Real-time processing enables immediate feedback, which is essential for applications like sports performance analysis or monitoring systems. The objective ensures latency is minimized so users receive timely and actionable information.
4. **Wireless Communication and User Interface:** One objective is to transmit data wirelessly to a mobile app, computer, or cloud platform. This allows remote monitoring and eliminates the need for wired connections, increasing portability and ease of use. The user interface will display data visually using charts, graphs, and alerts for easier interpretation. Wireless integration increases the system's applicability in real-world scenarios and enhances user experience.
5. **Testing, Evaluation, and Scalability:** After development, the system will be tested for accuracy, reliability, and stability under varying conditions. Performance metrics will be compared against standard benchmarks to quantify effectiveness. The system will be designed to allow scalability, enabling the addition of new sensors or features without redesigning the core system. This objective ensures long-term usability, adaptability, and relevance of the project. Evaluation provides measurable results that justify the success and applicability of the project.

2.4 Innovations and Contributions

Our project introduces several novel contributions beyond existing solutions:

- **Integration of Multiple Sensors:** The project innovatively combines sensors such as MPU6050 (accelerometer & gyroscope) and microphones into a single system. This enables simultaneous monitoring of motion, orientation, and sound, which is rarely addressed in conventional systems. The integrated approach provides more comprehensive and accurate analysis compared to isolated sensor systems.
- **Practical Application in Performance Analysis:** The project provides actionable insights for applications such as sports training, motion tracking, and environmental monitoring. By translating sensor data into measurable performance metrics, it bridges the gap between theoretical knowledge and practical usage. This innovation supports informed decision-making, continuous improvement, and skill enhancement.
- **Automation and Reduced Human Dependency:** By automating data collection and analysis, the system minimizes human errors and reduces manual workload. It ensures consistent monitoring over time, which is difficult to achieve with conventional manual systems. This contribution highlights the efficiency and reliability of automated systems in real-world scenarios.
- **Real-Time Wireless Monitoring:** The use of NodeMCU (ESP8266) allows real-time data transmission over Wi-Fi to mobile apps or cloud platforms. This reduces dependency on manual observation and enables remote monitoring, which is a significant enhancement over traditional setups. Users can access live metrics and receive instant feedback, improving efficiency and decision-making.
- **Compact and Cost-Effective Designs:** The system uses affordable, off-the-shelf components while maintaining functionality and accuracy. This innovation makes the solution accessible to educational institutes, amateur sports enthusiasts, or small-scale applications. Compact design ensures portability and ease of integration into existing environments.
- **Enhanced Accuracy and Reliability:** Calibration, noise filtering, and data processing algorithms improve measurement precision. This contributes to the reliability and consistency of the system under various operational conditions. Unlike conventional systems, the project reduces errors caused by manual handling or environmental interference.

2.5 Future Integration and Extensions

- **Integration with Mobile Applications:** The smart bat can be connected to a dedicated mobile app to provide real-time performance analytics. Users can track metrics such as swing speed, bat angle, impact force, and shot consistency over time. The app could store historical data, provide trend analysis, and suggest personalized improvement tips.
- **Cloud-Based Data Storage and Analysis:** Future versions can transmit data to cloud platforms for long-term storage and advanced analytics. Coaches and players could access performance data remotely, enabling detailed feedback and virtual training sessions. Machine learning algorithms could analyze large datasets to identify patterns and optimize training routines.
- **Integration with Wearable Devices:** The smart bat can be paired with wearables such as smart gloves or wristbands to capture additional motion and physiological data. This would provide a more holistic view of player performance, including swing mechanics and body posture. Combining wearable data with bat data could help prevent injuries by analyzing improper techniques.
- **Advanced Sensor Upgrades:** Future versions could incorporate additional sensors like impact vibration sensors, strain gauges, or pressure sensors. These sensors could provide more precise measurements of bat-ball contact, power, and timing. Improved sensor technology would enhance accuracy and provide deeper insights for performance improvement.
- **Gamification and Virtual Training:** Future versions could link the smart bat to virtual training environments or simulation games. Players could practice with interactive drills, improving engagement and motivation. Gamification could help younger players develop skills in a fun, competitive way while tracking progress.
- **Multi-Player Performance Comparison:** Data from multiple bats could be synchronized to compare performance across players. This could be used in academies or tournaments for benchmarking skills, tracking progress, and designing training programs. Leaderboards or analytics dashboards could encourage healthy competition and skill development.

Chapter 3

DESCRIPTION

3.1 Functional overview

The IoT-Driven Smart Bat for Real-Time Swing and Shot Evaluation is an intelligent sports analytics system designed to enhance player performance by continuously monitoring, analyzing, and providing instant feedback on batting technique using embedded sensors, edge processing, and cloud-based visualization. The system integrates inertial, acoustic, and wireless communication technologies within a cricket or baseball bat to capture fine-grained motion and impact data during gameplay or training sessions. At the core of the smart bat is a microcontroller unit such as NodeMCU or ESP32, which acts as the central processing and communication hub, interfacing with sensors including a multi-axis MPU6050 inertial measurement unit and a high-sensitivity microphone or piezoelectric impact sensor. The MPU6050 combines a 3-axis accelerometer and 3-axis gyroscope to measure linear acceleration, angular velocity, bat orientation, swing angle, speed, rotation, and follow-through dynamics in real time, while the microphone or vibration sensor detects the exact moment and intensity of ball impact. During operation, the bat continuously samples motion and sound data at a high frequency to ensure accurate capture of rapid swing events, and the raw sensor signals are filtered using digital filtering techniques such as low-pass or complementary filters to eliminate noise and sensor drift. Before active usage, the system performs an initial calibration phase in which the bat is held in a known rest position to establish baseline offset values for acceleration and angular velocity, ensuring consistency and reliability of subsequent measurements. As the player performs a swing, the system tracks the bat's movement path, angular displacement, and speed profile, identifying key phases such as backlift, downswing, impact, and follow-through.

The impact detection module analyzes sudden spikes in vibration and acoustic signals to precisely determine the instant of ball contact, and correlates this event with concurrent MPU6050 data to assess the bat's orientation and position at impact. Using predefined thresholds and machine-learned or rule-based classification models, the system categorizes the shot impact location as top edge, middle (sweet spot), or bottom edge, enabling

objective evaluation of shot quality. For enhanced training feedback, the system supports selectable batting modes such as forward defense, left drive, or right drive, allowing players to focus on specific stroke techniques; each mode has associated motion patterns and angular ranges stored in memory, against which live sensor readings are compared to label the stroke as correct or incorrect. A buzzer or haptic feedback unit embedded in the bat provides immediate tactile or audible feedback when an optimal center hit or correct swing is detected, reinforcing proper technique through real-time cues. Simultaneously, the microcontroller transmits processed data wirelessly over Wi-Fi using IoT protocols to a cloud platform or mobile application such as the Blynk dashboard, enabling real-time visualization of swing metrics including bat speed, swing angle, impact force, stroke type, and accuracy score.

Coaches and players can monitor live performance remotely, review historical session data, and analyze trends over time to identify strengths, weaknesses, and areas for improvement. The cloud backend stores timestamped swing data, enabling session-wise comparison, consistency analysis, and performance benchmarking. Advanced analytics modules can further compute derived parameters such as swing tempo, reaction time, impact efficiency, and energy transfer, providing deeper biomechanical insights. The system architecture emphasizes low latency and energy efficiency, ensuring that feedback is delivered instantly without disrupting the natural feel of the bat, while optimized power management allows prolonged usage on a compact rechargeable battery. From a functional perspective, the smart bat transforms traditional sports equipment into an intelligent cyber-physical system by seamlessly combining sensing, computation, connectivity, and user interaction. It reduces reliance on subjective coaching feedback by delivering quantifiable, objective performance metrics and supports data-driven training methodologies suitable for amateur players, professional athletes, and sports academies alike. The modular design allows scalability and future enhancements such as AI-based shot recognition, cloud-based machine learning models, video synchronization, and integration with wearable devices, making the IoT-Driven Smart Bat a comprehensive solution for real-time swing and shot evaluation in modern sports training ecosystems.

3.2 System Design

The system design of the IoT-Driven Smart Bat for Real-Time Swing and Shot Evaluation is structured around an integrated embedded sensing, processing, and communication framework that enables accurate capture and analysis of batting performance parameters in real time. The overall architecture consists of three major layers: the sensing layer, the processing and control layer, and the IoT communication and application layer, all of which work cohesively to deliver instantaneous feedback and long-term performance insights. The sensing layer is responsible for acquiring physical motion and impact data generated during bat swings and ball contact. This layer primarily includes an MPU6050 inertial measurement unit and an impact detection sensor such as a microphone or piezoelectric vibration sensor. The MPU6050 provides six degrees of freedom by combining a three-axis accelerometer and a three-axis gyroscope, allowing precise measurement of bat acceleration, angular velocity, orientation, swing angle, and rotational movement. The impact sensor detects high-frequency vibrations or acoustic peaks produced at the moment of ball contact, enabling accurate identification of impact timing and force. These sensors are mounted inside the bat in a manner that preserves balance and minimizes interference with the player's natural swing. The processing and control layer forms the core intelligence of the system and is implemented using a Wi-Fi-enabled microcontroller such as NodeMCU or ESP32. This unit continuously reads sensor data via I²C and analog interfaces, performs signal conditioning, and executes decision-making algorithms. Raw sensor outputs are filtered using digital filtering techniques to reduce noise and motion artifacts. A calibration routine is executed during system initialization to determine sensor offsets and baseline values, ensuring repeatable and accurate measurements. During operation, the microcontroller analyzes motion patterns to identify different swing phases including backlift, downswing, impact, and follow-through. By correlating impact sensor data with instantaneous orientation and acceleration values from the MPU6050, the system determines the quality of contact and classifies the shot as top edge, middle (sweet spot), or bottom edge.

1. Sensor Data Acquisition:

The smart bat is embedded with multiple sensors to capture physical parameters during play:

[1] MPU6050 (Accelerometer + Gyroscope)

- Measures bat orientation, angular velocity, and acceleration.
- Captures swing speed, bat angle, and motion patterns.

[2] Microphone / Impact Sensor

- Detects ball–bat contact through vibration or sound intensity.
- Helps determine impact timing and force.

2. Embedded Processing Unit:

NodeMCU (ESP8266) acts as the core controller

Functions:

- Reads raw sensor data via I²C / ADC
- Filters noise using averaging or low-pass filtering
- Synchronizes motion and impact events

3. Calibration & Reference Setup:

The system performs **one-time calibration**:

- Bat held in rest position
- Baseline MPU6050 values stored as reference

Ensures accuracy across different players and grip styles.

4. Swing & Impact Analysis:

During a shot, the system processes data in real time:

- Impact Detection
- Sudden spike in vibration/sound confirms ball contact

5. IoT Connectivity & Cloud Integration:

- NodeMCU sends processed data via Wi-Fi
- Connected to Blynk / IoT dashboard
- Displays:
 - Impact type
 - Swing mode

6. Real-Time Visualization & Coaching Support:

- Coaches receive instant feedback remotely
- Enables:
 - 1) Objective performance tracking
 - 2) Data-driven coaching decisions
 - 3) Skill improvement over time

7. Feedback Mechanism:

- Buzzer / Haptic Feedback
- Triggered instantly for correct or sweet-spot hits
- Helps players self-correct without stopping practice

8. Overall System Benefits:

- Real-time performance evaluation.
- Portable and non-intrusive design.
- Low-cost IoT solution for sports analytics.
- Useful for training, skill assessment, and talent identification.

3.2.1 Block Diagram

Block Diagram:

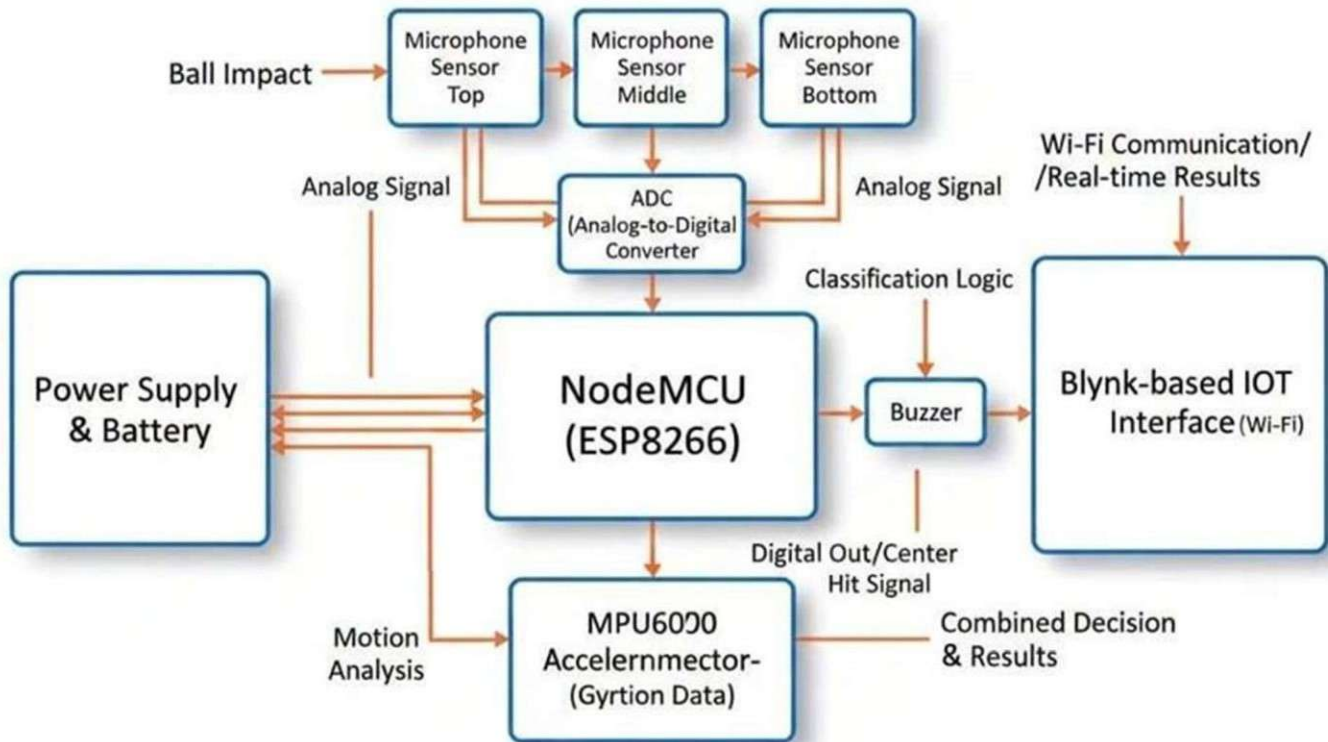


Figure 3.1. Block Diagram

3.3 Development Process

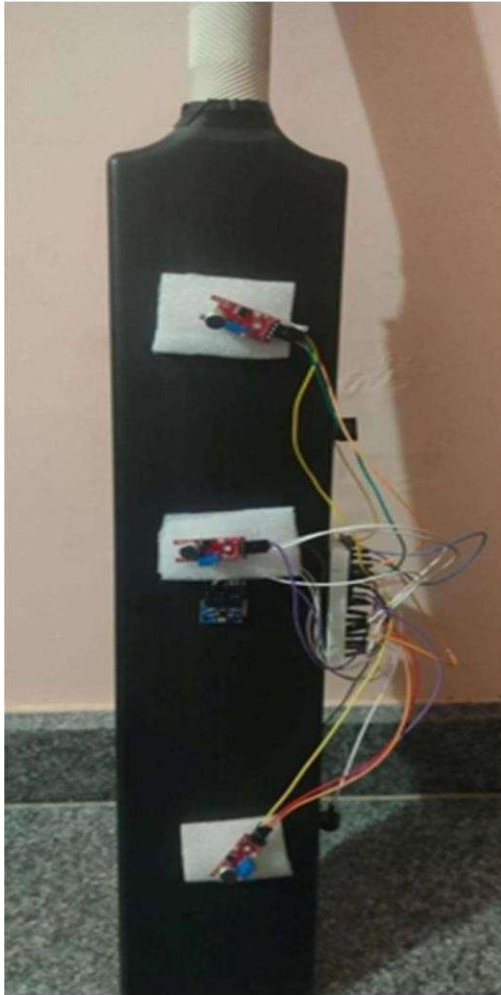


Figure 3.1

This **side model of the IoT-driven smart bat** is made by carefully fixing the sensing components along the face of the bat to accurately capture impact and motion during play. Multiple **impact or vibration sensors (microphone/sound sensors)** are mounted at different positions on the bat surface using adhesive pads or tape, allowing the system to distinguish between **top edge, middle (sweet spot), and bottom edge** ball contacts. These sensors are connected through jumper wires routed neatly along the bat to the main control unit mounted on the other side. An **MPU6050 accelerometer and gyroscope module** is also placed strategically to sense bat orientation, swing angle, and movement patterns.

Proper spacing and firm attachment ensure reliable readings while maintaining the bat's balance. This side-mounted arrangement helps the system correlate impact location with swing data, enabling precise real-time shot evaluation and performance analysis. evaluation.

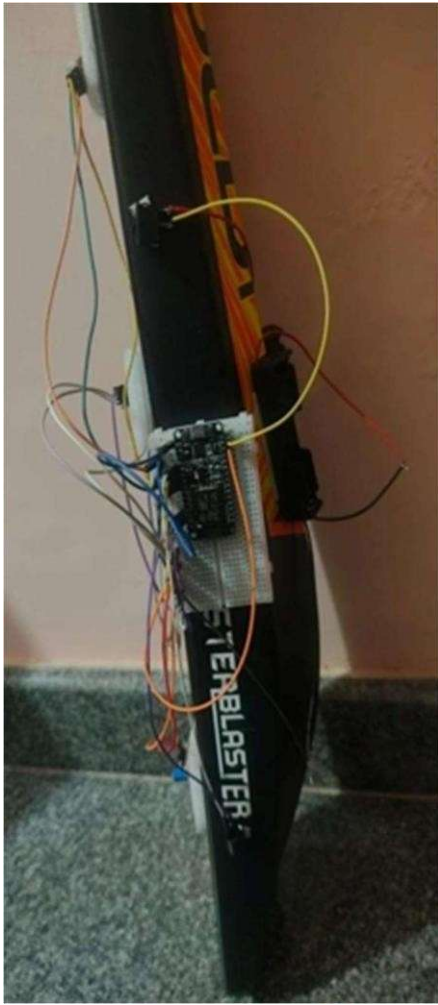


Figure 3.2

This model of the **IoT-Driven Smart Bat** is made by mounting all the required electronic components directly onto a cricket bat in a compact and non-intrusive way. A **NodeMCU (ESP8266)** microcontroller is fixed on a small **breadboard**, which is firmly attached to the bat handle/body using tape or clamps.

An **MPU6050 accelerometer and gyroscope sensor** is placed along the bat to accurately capture swing motion, angle, and speed, while a **microphone or impact sensor** is positioned to sense vibrations or sound generated when the ball hits the bat. All components are interconnected using jumper wires, ensuring proper power supply and data communication through I²C and analog pins.

The system is powered by a **battery**, making it portable for real-time practice sessions. Once assembled, the microcontroller is programmed to read sensor data, process swing and impact information, and transmit results wirelessly via Wi-Fi to a mobile or cloud dashboard, enabling real-time swing and shot evaluation.

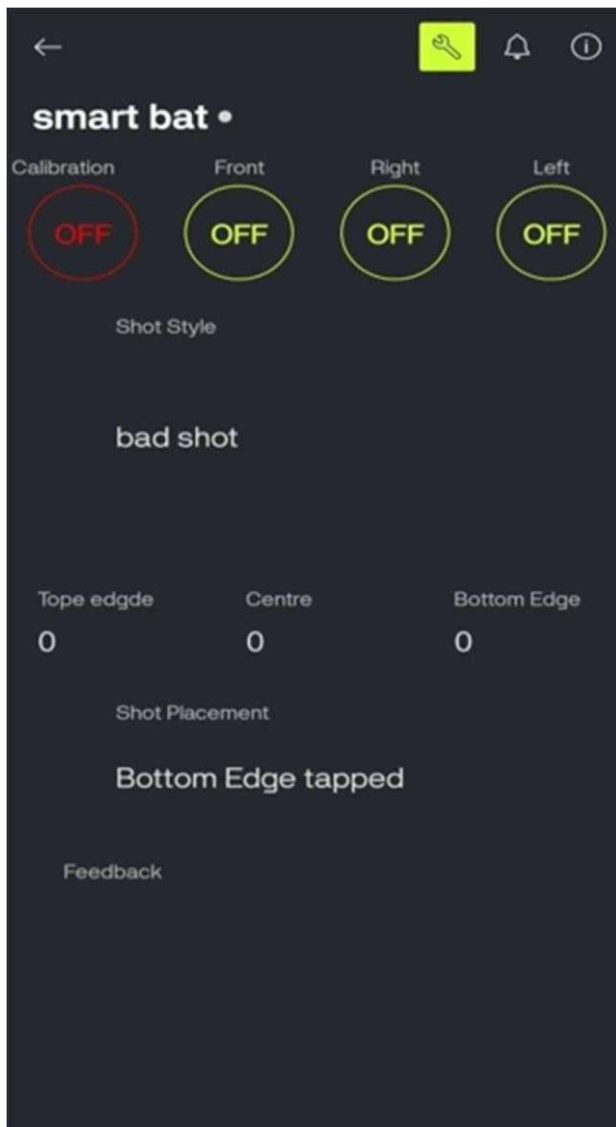


Figure 3.3

This picture shows the **mobile dashboard interface of the IoT-based Smart Bat system**, used to monitor and analyze batting performance in real time. At the top, the **calibration and shot-direction modes** (Front, Right, Left) are displayed as circular indicators, which remain **OFF**, indicating that no mode is currently active or calibration has not been enabled.

The **Shot Style** section displays the result of the most recent swing, where **“bad shot”** indicates that the swing or bat angle did not match the predefined correct motion pattern. Below this, the **impact count section** shows values for **Top Edge, Centre, and Bottom Edge**, all currently at zero, which represent the number of times the ball has hit each region of the bat.

The **Shot Placement** message, **“Bottom Edge tapped”**, indicates that the system detected ball contact near the lower edge of the bat during the last attempt. Finally, the **Feedback** area is reserved for real-time alerts or coaching suggestions, which can be updated dynamically based on performance. Overall, this interface provides instant visual feedback to the player and coach, helping in identifying mistakes and improving shot execution.

3.3.1 Hardware Setup

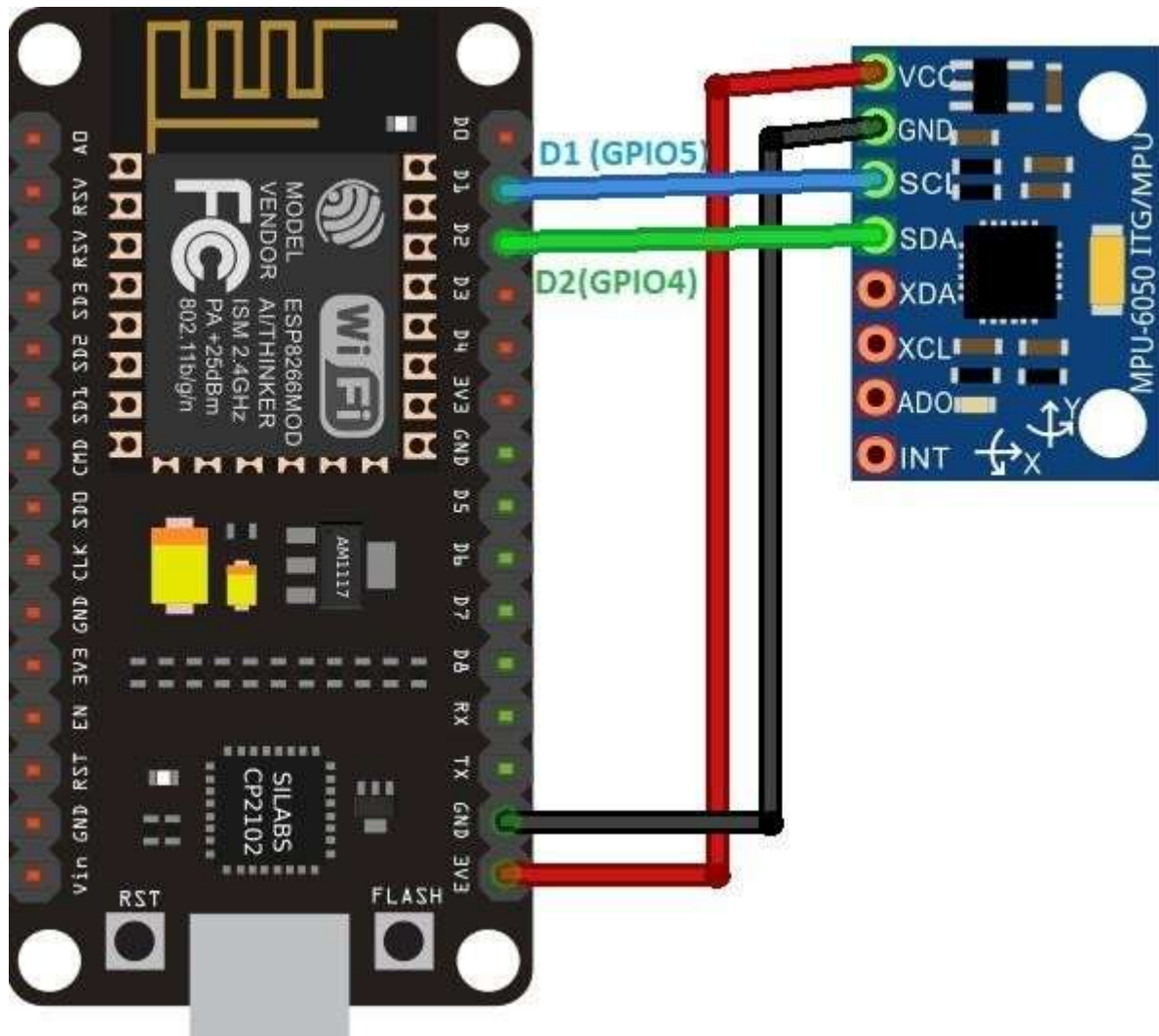


Figure 3.4. Pin Diagram

"This picture shows the hardware connection diagram between the NodeMCU (ESP8266) and the MPU6050 accelerometer–gyroscope sensor, which is used in the IoT-Driven Smart Bat system to measure swing motion and orientation. The VCC pin of the MPU6050 is connected to the 3.3 V pin of the NodeMCU to power the sensor, while GND is connected to GND to complete the common ground. Communication between the two devices is done using the I²C protocol, where the SCL (clock) pin of the MPU6050 is connected to D1 (GPIO5) of the NodeMCU, and the SDA (data) pin is connected to D2 (GPIO4). These two lines allow continuous transmission of acceleration and gyroscope data from the sensor to the microcontroller. The remaining pins such as INT, AD0, XDA, and XCL are not used in this setup. With this wiring, the NodeMCU can reliably read real-time motion data from the MPU6050, which is essential for analyzing bat swing angle, speed, and direction in the smart bat system.

Chapter 4

RESULT AND ANALYSIS

4.1. Overview Of Outcomes

The experimental setup plays a crucial role in validating the effectiveness, accuracy, and reliability of any embedded system-based project. In the case of the Smart Bat System, the experimental setup was designed carefully to simulate real-world cricket practice conditions while ensuring controlled testing for accurate result analysis. The objective of this setup was not only to test whether the system works but also to evaluate its consistency, responsiveness, robustness, and suitability for practical training environments. Since cricket is a highly dynamic sport involving rapid movements, vibrations, and directional changes, the experimental configuration had to account for mechanical, electrical, and environmental factors simultaneously.

The Smart Bat experimental setup consisted of both hardware components and software infrastructure, integrated seamlessly to operate as a single intelligent system. The primary hardware elements included a standard wooden cricket bat, three microphone-based vibration sensors, an MPU6050 accelerometer–gyroscope module, a NodeMCU (ESP8266) microcontroller, a buzzer unit, and a portable power supply in the form of a lithium-ion battery. These components were mounted securely on the bat using a compact enclosure to ensure minimal disturbance to the natural balance and handling of the bat. The experimental setup also incorporated a smartphone running the Blynk IoT application, which served as the real-time user interface for monitoring system output.

A critical requirement of the experimental setup was that it should not alter the batting technique of the player. Therefore, special attention was given to the placement of sensors and electronic components. The enclosure was positioned near the lower portion of the bat handle, where it would have minimal effect on swing dynamics. The microphone sensors were placed at three distinct regions along the bat blade—top edge, center (sweet spot), and bottom edge. This placement was selected based on vibration propagation characteristics observed during preliminary trials. When a cricket ball strikes the bat, vibrations travel along the blade, and the sensor closest to the impact point experiences the highest amplitude.

The MPU6050 motion sensor was installed inside the enclosure to ensure protection from mechanical shocks while maintaining accurate orientation measurement. The sensor was aligned carefully with the bat's longitudinal axis to ensure correct interpretation of forward, left, and right swing motions. Any misalignment could lead to erroneous angle calculations and false shot classification. Hence, alignment calibration was performed before initiating experimental trials. The NodeMCU microcontroller served as the central processing unit, collecting sensor data, executing classification algorithms, and transmitting results wirelessly via Wi-Fi to the Blynk cloud server.

To power the system, a rechargeable lithium-ion battery was used, providing a stable 3.3V regulated supply to all components. Power stability was a major consideration in the experimental setup because voltage fluctuations could affect sensor readings, especially microphone sensors that are sensitive to noise. Voltage regulation circuitry was tested extensively before experimentation to ensure consistent performance during prolonged usage. The buzzer module was connected to a digital output pin of the NodeMCU and positioned such that the sound feedback was audible to the player during practice sessions without being distracting.

On the software side, the experimental setup required proper configuration of the Arduino IDE, Blynk IoT platform, and Wi-Fi credentials. The firmware uploaded to the NodeMCU was designed to initialize all sensors, perform calibration routines, continuously monitor sensor data, and respond to events such as ball impact and swing motion. The Blynk application was configured with multiple virtual buttons and display widgets, enabling mode selection, real-time feedback display, and system status monitoring. Each virtual pin was mapped precisely to a corresponding function in the firmware, ensuring synchronized communication between hardware and software layers.

Before conducting actual batting experiments, the system underwent a calibration phase, which is a vital part of the experimental setup. During calibration, the bat was held in a neutral position, and baseline readings of the MPU6050 were recorded. This step compensated for sensor offsets, environmental drift, and minor mounting imperfections. Similarly, microphone sensor thresholds

were adjusted based on ambient noise levels to prevent false triggering due to wind, ground vibrations, or hand movements. Calibration ensured that all subsequent experimental readings were relative to a stable reference point.

The experimental environment was chosen carefully to reflect typical cricket practice conditions while minimizing uncontrolled disturbances. Tests were conducted in an open practice area with sufficient space for natural batting movements. Both indoor and outdoor trials were conducted to observe system behavior under varying acoustic and vibration conditions. For outdoor testing, environmental factors such as wind and background noise were monitored, as these could influence microphone sensor readings. Indoor testing allowed controlled evaluation of sensor response without external interference.

To ensure the reliability of results, multiple trials were conducted for each type of shot and impact location. Forward shots, right shots, and left shots were performed repeatedly by the player while selecting the corresponding mode in the Blynk application. Similarly, controlled impacts were generated intentionally near the top edge, center, and bottom edge of the bat to validate impact detection accuracy. Each trial was observed, recorded, and compared with the system's output displayed on the Blynk interface. This repetitive methodology helped in identifying patterns, inconsistencies, and potential sources of error.

Another important aspect of the experimental setup was user interaction testing. Since the Smart Bat System is intended for real-world usage by players of different skill levels, the setup was evaluated for ease of use, responsiveness, and clarity of feedback. The system was tested by users with minimal technical knowledge to ensure that the feedback messages displayed on the Blynk app were intuitive and understandable. The buzzer feedback was also evaluated to determine whether it provided effective reinforcement for correct shots without causing distraction.

In summary, the experimental setup for the Smart Bat System was designed to closely emulate real-world cricket training conditions while maintaining controlled testing parameters for accurate performance evaluation. The careful integration of hardware components, software configuration, calibration procedures, environmental considerations, and safety measures ensured that the system could be tested reliably and repeatedly. This comprehensive experimental setup formed the foundation for obtaining meaningful results, which are discussed in the subsequent sections of this chapter.

4.2. HARDWARE IMPLEMENTATION

- The hardware implementation of the Smart Bat System represents the physical realization of the conceptual design discussed in the previous chapters. This stage of the project focuses on translating theoretical system architecture into a functional, reliable, and robust prototype capable of operating under real-world cricket training conditions. Hardware implementation is a critical phase because the performance of the entire system heavily depends on sensor placement accuracy, electrical stability, mechanical integrity, and seamless integration of all components. Any flaw in hardware implementation can directly affect system accuracy, reliability, and usability, irrespective of how well the software algorithms are designed.
- The Smart Bat hardware implementation was carried out by integrating multiple sensing modules, a microcontroller unit, feedback components, and a power supply system into a compact and portable assembly. The primary challenge during implementation was to ensure that the electronic components did not interfere with the natural handling, weight distribution, and swing dynamics of the cricket bat. Since cricket batting involves rapid acceleration, deceleration, and vibration, the hardware needed to withstand repeated mechanical stress while maintaining consistent electrical performance.
- The NodeMCU (ESP8266) microcontroller was chosen as the central hardware unit due to its compact size, low power consumption, and built-in Wi-Fi capability. During hardware implementation, the NodeMCU was mounted securely inside the enclosure using insulating spacers to prevent short circuits and mechanical damage. All sensor connections were routed carefully to the microcontroller's GPIO and I²C pins according to the predefined circuit design. Particular attention was paid to grounding and power routing to minimize electrical noise, which could adversely affect sensor readings, especially from microphone-based vibration sensors.
- Three microphone sensors were implemented as the primary impact detection units. These sensors were fixed along the length of the bat blade at three distinct locations corresponding to the top edge, center (sweet spot), and bottom edge. The positioning of these sensors was determined through preliminary testing and vibration propagation

analysis. When a cricket ball impacts the bat, the vibration energy is not uniformly distributed; instead, it attenuates with distance from the point of impact. By placing sensors at strategic locations, the system can reliably determine which region of the bat experienced the highest vibration amplitude. During hardware implementation, the sensors were attached firmly using adhesive material to ensure direct contact with the bat surface, thereby improving vibration transmission and reducing damping effects.

- The MPU6050 accelerometer–gyroscope module was implemented as the motion sensing unit of the system. This sensor was mounted inside the enclosure to protect it from direct mechanical shocks while ensuring stable orientation measurement. The alignment of the MPU6050 with respect to the bat's longitudinal axis was a crucial consideration during implementation. Improper alignment could lead to incorrect interpretation of swing direction and angular displacement. Therefore, the sensor was mounted parallel to the bat handle axis, and its orientation was verified during calibration. Shielded wiring was used for I²C communication lines to minimize interference and ensure reliable data transmission between the MPU6050 and the NodeMCU.
- A buzzer module was incorporated into the hardware implementation to provide immediate auditory feedback to the player. The buzzer was connected to a digital output pin of the NodeMCU and placed within the enclosure in such a way that the sound produced was clearly audible during batting practice. The buzzer activation duration was controlled through firmware to ensure that feedback was noticeable but not distracting. This component played an important role in reinforcing correct batting technique by providing instant confirmation of sweet-spot hits.
- Power management was another critical aspect of hardware implementation. A rechargeable lithium-ion battery was used to power the entire system, ensuring portability and ease of use. The battery was connected to a voltage regulation circuit that provided a stable 3.3V supply to the NodeMCU and sensors. Voltage stability was verified using a multimeter under both idle and active conditions. The implementation ensured that sudden current draws during Wi-Fi transmission or buzzer activation did not cause voltage drops that could reset the microcontroller or distort sensor readings. Proper insulation and protective casing were used to prevent accidental short circuits and ensure user safety.

- The physical enclosure played a vital role in the overall hardware implementation. A compact enclosure was designed to house all electronic components securely. The enclosure was selected based on criteria such as lightweight construction, mechanical strength, and ease of attachment to the bat. It was mounted near the lower portion of the bat handle to minimize its effect on swing balance. Ventilation openings were provided to prevent heat accumulation, and internal padding was added to absorb vibrations and protect sensitive components. The enclosure design ensured that the system remained functional even after repeated batting sessions.
- Wiring and interconnections were implemented using flexible jumper wires and soldered joints where necessary. All connections were strain-relieved to prevent loosening due to continuous vibrations during bat swings. Cable management inside the enclosure was carefully arranged to avoid wire entanglement or pressure on components. After assembly, continuity tests were conducted to verify correct connections and eliminate wiring faults.
- Once the complete hardware assembly was implemented, the system underwent a series of validation checks before experimental testing. These checks included power-on verification, sensor response testing, Wi-Fi connectivity testing, and buzzer activation testing. Each sensor's output was monitored using serial communication to ensure that raw data values were within expected ranges. Any anomalies observed during initial testing were addressed through minor adjustments in sensor placement or wiring.
- Overall, the hardware implementation of the Smart Bat System successfully transformed the conceptual design into a practical, real-world prototype. The careful selection of components, precise sensor placement, robust power management, and secure mechanical assembly ensured reliable system performance. This hardware foundation enabled accurate data acquisition and real-time feedback, which are essential for evaluating the effectiveness of the Smart Bat System. The results obtained from this implementation are analyzed and discussed in subsequent sections of this chapter.

4.3 BLYNK OUTPUT AND RESULT OBSERVATION

- The Blynk IoT platform serves as the primary user interface for the Smart Bat System, enabling real-time visualization of sensor outputs and system decisions. This section focuses on the observed results displayed on the Blynk application during experimental trials and explains how these outputs validate the functionality of the system. The integration of Blynk provides an interactive and user-friendly medium through which players and coaches can instantly understand shot quality without requiring technical knowledge.
- During system operation, the Blynk application displays multiple outputs corresponding to impact detection, shot direction classification, and system status messages. Virtual buttons are used to select different training modes such as forward shot mode, right shot mode, and left shot mode. Once a mode is selected, the system continuously monitors motion data from the MPU6050 sensor and impact signals from the microphone sensors. Upon detecting a batting event, the processed result is transmitted wirelessly to the Blynk cloud server and reflected instantly on the mobile interface.
- The impact detection results are displayed in textual form, indicating whether the ball made contact with the top edge, center (sweet spot), or bottom edge of the bat. When a center hit is detected, the system simultaneously activates the buzzer and displays a confirmation message on the Blynk interface. This dual feedback mechanism ensures that the player receives both auditory and visual confirmation of a correct shot. For top edge or bottom edge impacts, the application displays appropriate warning messages, allowing the player to recognize timing errors immediately.
- Shot direction results are also displayed on the Blynk dashboard based on the selected mode. If the detected bat motion matches the selected training mode, the application displays a message indicating a correct shot. In cases where the motion does not match the selected mode, the system displays a “Bad Shot” or “Incorrect Direction” message. Additionally, the system handles mode conflicts by displaying an error message when more than one mode is selected simultaneously, ensuring structured and disciplined practice.
- Throughout multiple experimental trials, the Blynk outputs remained consistent, responsive, and easy to interpret. The real-time display of results confirmed that the Smart Bat System successfully integrates sensor data processing with IoT-based feedback. These observations demonstrate that the Blynk interface effectively bridges the gap between embedded hardware and user interaction, making the system suitable for practical cricket training applications.

4.4 PERFORMANCE ANALYSIS

Performance analysis is an essential aspect of evaluating the effectiveness and reliability of the Smart Bat System. This section analyzes the system's performance in terms of accuracy, response time, stability, and usability based on the experimental observations recorded during testing. The goal of this analysis is to determine whether the system meets the intended objectives and performs consistently under practical conditions.

The accuracy of the Smart Bat System was evaluated by comparing actual shot conditions with the system's detected outputs. Impact detection accuracy was found to be high, as the microphone sensors reliably identified the region of ball contact on the bat. The differentiation between top edge, center, and bottom edge impacts was consistent across multiple trials, indicating effective sensor placement and threshold calibration. Similarly, shot direction classification using the MPU6050 sensor showed reliable results when the bat motion closely followed predefined directional patterns.

Response time is a critical performance parameter in real-time training systems. The Smart Bat System demonstrated low latency between shot execution and feedback display on the Blynk application. The total response time, including sensor data acquisition, processing, and IoT transmission, was observed to be within a fraction of a second. This ensured that feedback was delivered almost instantly, allowing players to correlate their actions with system output effectively.

System stability was assessed during extended testing sessions. The hardware operated reliably without unexpected resets or communication failures, provided that Wi-Fi connectivity remained stable. Power consumption was found to be within acceptable limits, enabling multiple practice sessions on a single battery charge. Minor sensor drift observed during prolonged use was mitigated through periodic calibration.

Overall, the performance analysis confirms that the Smart Bat System operates accurately, responsively, and reliably. The system meets the functional requirements of real-time cricket shot analysis and demonstrates suitability for both training and educational purposes.

4.5 DISCUSSION

This section discusses the significance of the experimental results obtained from the Smart Bat System and interprets their implications in the context of cricket training and sports analytics. The discussion aims to relate the observed system performance to the objectives defined earlier and highlight the practical usefulness of the proposed solution.

The experimental results clearly indicate that the Smart Bat System successfully addresses key limitations of traditional cricket coaching methods. The accurate detection of impact location helps players understand timing errors, while motion-based shot classification assists in correcting swing direction. The combination of these two feedback mechanisms provides a comprehensive evaluation of batting performance, which is rarely available in low-cost training tools.

The use of an IoT platform such as Blynk significantly enhances the usability of the system. Real-time feedback enables players to make immediate adjustments to their technique, thereby accelerating the learning process. Additionally, the system promotes self-training by reducing dependence on continuous coach supervision. This is particularly beneficial for beginners and amateur players who may not have access to professional coaching facilities.

While the system demonstrated strong performance, certain limitations were observed, such as sensitivity to Wi-Fi connectivity and the need for periodic calibration. However, these limitations do not undermine the overall effectiveness of the system and can be addressed through future enhancements. The results validate the design approach and confirm that integrating embedded systems with IoT technology can create practical and affordable sports training solutions.

In summary, the discussion reinforces that the Smart Bat System achieves its intended goals and has the potential to significantly improve cricket training methodology when adopted at grassroots and institutional level

Chapter 5

CONCLUSION

5.1 Key Learnings

The Smart Bat System was successfully designed, implemented, and tested with the objective of enhancing cricket training through the integration of embedded systems and IoT technology. Traditional cricket coaching methods largely depend on human observation, which often fails to capture minute errors related to timing, impact location, and swing direction. This project addresses these limitations by providing an intelligent, real-time feedback system capable of objectively evaluating batting performance during practice sessions.

The system effectively combines microphone-based vibration sensors and an MPU6050 accelerometer–gyroscope module to analyze both ball impact location and bat motion. The microphone sensors enable accurate detection of top edge, center, and bottom edge impacts, allowing players to understand timing-related mistakes instantly. The motion sensor tracks bat orientation and swing direction, making it possible to classify different shot types based on player-selected training modes. The integration of these sensing mechanisms ensures a comprehensive evaluation of batting technique rather than focusing on a single performance parameter.

The NodeMCU (ESP8266) microcontroller played a crucial role in processing sensor data and enabling wireless communication with the Blynk IoT platform. The real-time display of results on the mobile application, along with auditory feedback through the buzzer, made the system intuitive and user-friendly. Experimental testing confirmed that the system delivers fast response times, reliable operation, and consistent results under practical conditions. The structured mode-based training approach further improved accuracy by preventing ambiguous shot interpretation.

Overall, the Smart Bat System demonstrates that affordable and compact embedded solutions can significantly improve sports training methodologies. The project

successfully meets its objectives and proves that IoT-based sports analytics can be made accessible to beginners, students, and amateur players without the need for expensive professional equipment. This work contributes to the growing field of smart sports technology and highlights the potential of interdisciplinary engineering applications in real-world.

5.2 FUTURE SCOPE

Although the Smart Bat System performs effectively in its current form, several enhancements can be implemented to further improve functionality, accuracy, and scalability. One major area of future development involves the integration of machine learning algorithms. By collecting and analyzing large datasets of batting strokes, intelligent models can be trained to classify a wider variety of shots and provide personalized feedback based on individual player performance patterns.

Another promising extension is the calculation of advanced performance metrics such as swing speed, impact force, and bat angle variation. These parameters can offer deeper insights into a player's technique and help in professional-level performance analysis. Additionally, cloud-based data storage and web dashboards can be introduced to maintain training history, generate performance graphs, and enable remote coaching for academies and institutions.

The system can also be expanded by integrating additional wearable sensors, such as smart gloves or foot-placement sensors, to provide a more holistic analysis of a player's biomechanics.

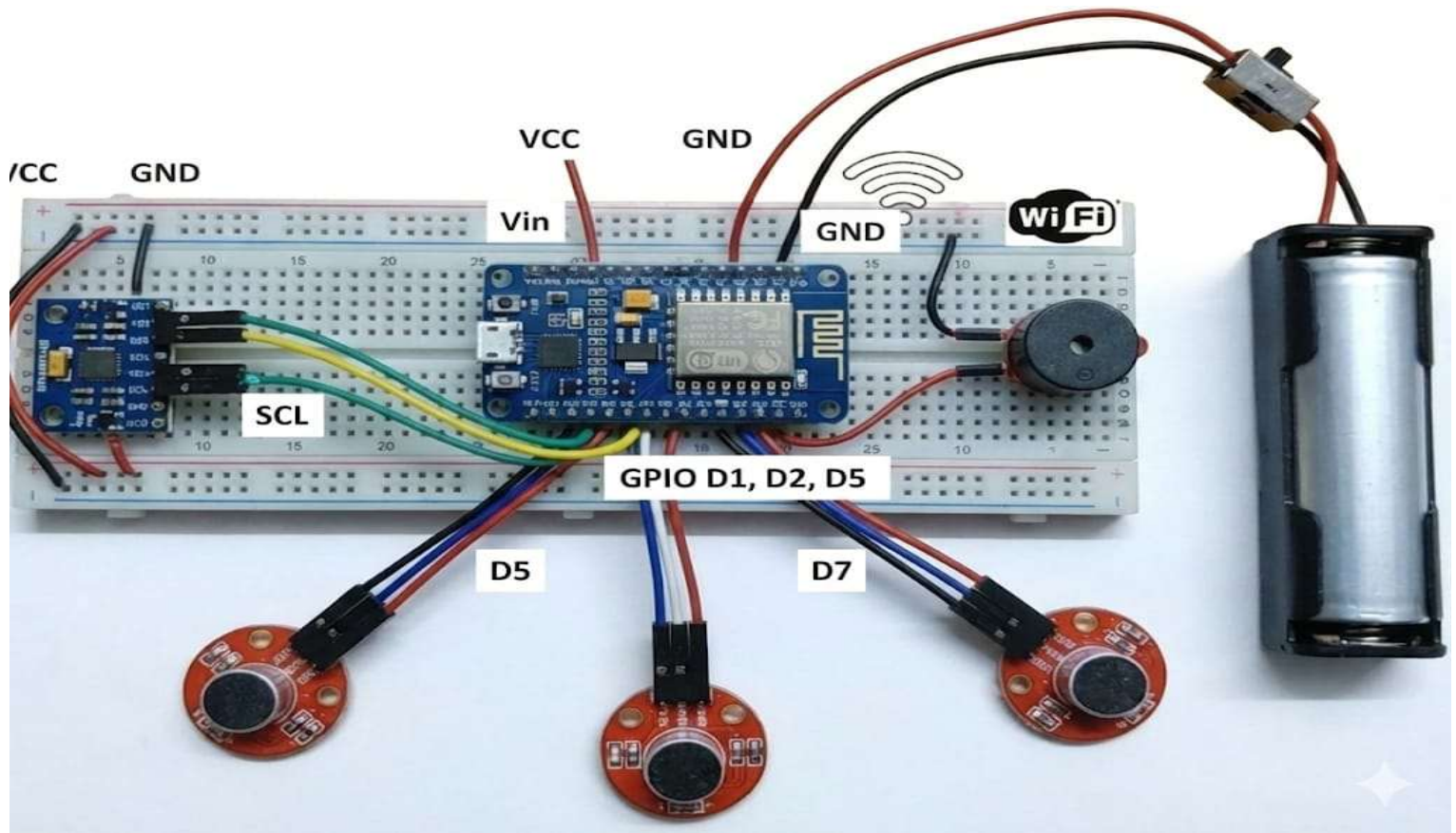
In the future, the Smart Bat System can evolve into a comprehensive smart training ecosystem capable of supporting multiple players, advanced analytics, and real-time coaching support. These enhancements would further strengthen its role as a practical, low-cost, and innovative so applicability across diverse environments. Its compact, user-friendly design also ensures that players from all age groups can use it with minimal training.

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APPENDIX A

1. CIRCUIT DIAGRAM



2. COMPONENTS LIST

- NodeMCU (ESP8266) Wi-Fi enabled microcontroller
- Microphone-based vibration sensors (three units)
- MPU6050 accelerometer and gyroscope sensor module
- Buzzer module for auditory feedback
- Rechargeable lithium-ion battery
- Voltage regulator module (3.3 V)

CODE

```

#define BLYNK_TEMPLATE_ID "TMPL3yfx8TIqU"

#define BLYNK_TEMPLATE_NAME "smart bat"

#define BLYNK_AUTH_TOKEN "PZWBnKxazhQYWJYU00Cvao6Qp4zPejMt"


#define BLYNK_PRINT Serial


/***** LIBRARIES *****/

#include <Wire.h>

#include <MPU6050.h>

#include <ESP8266WiFi.h>

#include <BlynkSimpleEsp8266.h>


/***** WIFI *****/

char ssid[] = "Airtel_sugu_0550";
char pass[] = "Air@23860";


/***** MPU6050 *****/

MPU6050 mpu;


/***** VIRTUAL PINS *****/

const int RESULT_PIN = V3; // existing result
const int MIC_V_TOP = V6; // mic top -> V6
const int MIC_V_CENTER = V7; // mic centre -> V7
const int MIC_V_BOT = V8; // mic bottom -> V8
const int MIC_V_STRING = V9; // human readable event -> V9
const int FEEDBACK_VPIN = V10; // new: show random feedback when bad shot

```

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```

/***** PARAMETERS *****/

const int CAL_SAMPLES_REST = 200;

const int SAMPLE_DELAY_MS = 20;

const int MOVING_BUF = 10;

const int WINDOW_SIZE = 12;


const int MOVEMENT_MIN_GX = 350;

const unsigned long TRIGGER_COOLDOWN = 1500;


/***** ZONE TUNING (NARROWER / NON-OVERLAPPING) *****/

const int ZONE_THRESHOLD = 450; // reduced radius for left/right (narrower zones)

const int ZONE_DEADBAND = 150; // central dead zone to avoid ambiguous results


/***** NEW VERTICAL THRESHOLDS *****/

const int Y_LIFT_MIN = 800; // right shots → bat up

const int Z_LIFT_MIN = 600;

const int Y_DROP_MIN = 800; // left sweep → bat down

const int Z_DROP_MIN = 500;


/***** MICROPHONE INPUTS (hardware pins) *****/

const int MIC_TOP_PIN = D5; // GPIO14

const int MIC_CENTER_PIN = D6; // GPIO12

const int MIC_BOT_PIN = D7; // GPIO13


const unsigned long MIC_DEBOUNCE_MS = 50; // debounce for mic taps

const int buzzer = D8;


/***** FEEDBACK MESSAGES *****/

// 10 different feedback strings — edit these as you like

const char* feedbacks[] = {
```

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```
"Correct the angle",
"Play in another phase",
"Lower the bat downward",
"Step forward more",
"Keep elbow bent",
"Open the face less",
"Shift weight earlier",
"Follow through more",
"Bring bat down quicker",
"Watch ball earlier"
};

const int FEEDBACK_COUNT = sizeof(feedbacks)/sizeof(feedbacks[0]);

/***** GLOBALS *****/

float rest_mean_x = 0;
float rest_mean_y = 0;
float rest_mean_z = 0;

int16_t bufX[MOVING_BUF] = {0}, bufY[MOVING_BUF] = {0}, bufZ[MOVING_BUF] = {0};
int bufIndex = 0;
long sumX=0,sumY=0,sumZ=0;
bool bufFilled=false;

unsigned long lastTrigger=0;

bool sendEnabled=false;
bool rightMode=false;
bool leftMode=false;

/* mic state tracking */
```

IoT-Driven Smart Bat for Real-Time Swing and Shot Evaluation

```
int lastMicState[3] = {LOW, LOW, LOW}; // top, center, bottom
unsigned long lastMicTime[3] = {0,0,0};

/* feedback state */

int lastFeedbackIndex = -1; // -1 means no feedback shown currently
bool lastShotWasGood = true; // track whether previous recognized shot was "good"

/***** BLYNK HANDLERS *****/

BLYNK_WRITE(V1){
  if(param.asInt()==1){
    calibrateRest();
    Blynk.virtualWrite(V1,0);
  }
}

BLYNK_WRITE(V2){ sendEnabled = param.asInt(); }
BLYNK_WRITE(V4){ rightMode = param.asInt(); }
BLYNK_WRITE(V5){ leftMode = param.asInt(); }

/***** SETUP *****/

void setup(){
  Serial.begin(115200);
  delay(100);

  pinMode(buzzer, OUTPUT);
  digitalWrite(buzzer, LOW);

  // seed PRNG from A0 (floating/noise) for random picks
  randomSeed(analogRead(A0));

  // I2C for MPU (SDA, SCL)
```

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```
Wire.begin(D2, D1); // D2 = SDA, D1 = SCL

delay(10);

// Initialize MPU and check connection
mpu.initialize();
delay(50);
Serial.println("Checking MPU6050 connection...");
unsigned long start = millis();
while (!mpu.testConnection()) {
    Serial.println("MPU6050 not detected - retrying...");
    delay(500);
    if (millis() - start > 5000) {
        Serial.println("Still waiting for MPU6050... make sure wiring/power/I2C are correct.");
        start = millis();
    }
}
Serial.println("MPU6050 detected and ready.");
delay(50); // give MPU a moment

// mic pins
pinMode(MIC_TOP_PIN, INPUT);
pinMode(MIC_CENTER_PIN, INPUT);
pinMode(MIC_BOT_PIN, INPUT);

// ensure smoothing buffers cleared
for (int i=0;i<MOVING_BUF;i++){
    bufX[i]=bufY[i]=bufZ[i]=0;
}
sumX = sumY = sumZ = 0;
bufIndex = 0;
```

IoT-Driven Smart Bat for Real-Time Swing and Shot Evaluation

```
bufFilled = false;

// connect Blynk
Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);

// initial read of mic states
lastMicState[0] = digitalRead(MIC_TOP_PIN);
lastMicState[1] = digitalRead(MIC_CENTER_PIN);
lastMicState[2] = digitalRead(MIC_BOT_PIN);
for(int i=0;i<3;i++) lastMicTime[i] = millis();

// initial calibration of resting gyro means
calibrateRest();

// ensure feedback starts blank
Blynk.virtualWrite(FEEDBACK_VPIN, "");
lastFeedbackIndex = -1;
lastShotWasGood = true;
}

/***** ZONE CLASSIFIER (exclusive, narrow zones + deadband) *****/
int classifyZone(float dX){
    if (fabs(dX) <= ZONE_DEADBAND) return 0; // center / too small -> middle
    if (dX <= -ZONE_THRESHOLD) return 1; // LEFT
    if (dX >= ZONE_THRESHOLD) return 2; // RIGHT
    return 0;
}

/***** MAIN LOOP *****/
void loop(){
```

```
Blynk.run();

// check microphones first (fast)
checkMic(0, MIC_TOP_PIN, MIC_V_TOP); // top
checkMic(1, MIC_CENTER_PIN, MIC_V_CENTER); // centre
checkMic(2, MIC_BOT_PIN, MIC_V_BOT); // bottom

// read gyro (rotation) values
int16_t gx = mpu.getRotationX();
int16_t gy = mpu.getRotationY();
int16_t gz = mpu.getRotationZ();

updateBuffer(gx, gy, gz);
float avgX = getAvgX();
float avgY = getAvgY();
float avgZ = getAvgZ();

float deltaX = avgX - rest_mean_x;
float deltaY = avgY - rest_mean_y;
float deltaZ = avgZ - rest_mean_z;

bool modeError = (leftMode && rightMode);
if(modeError){
    send("MODE ERROR");
    return;
}

// Quick swing check to avoid expensive window sampling for small movements
bool swing = fabs(deltaX) > (MOVEMENT_MIN_GX * 0.5);
if(!swing) return;
```


IoT-Driven Smart Bat for Real-Time Swing and Shot Evaluation

```
if(millis() - lastTrigger < TRIGGER_COOLDOWN) return;

long sx=0, sy=0, sz=0;

for(int i=0;i<WINDOW_SIZE;i++){
    int16_t x = mpu.getRotationX();
    int16_t y = mpu.getRotationY();
    int16_t z = mpu.getRotationZ();
    sx += x; sy += y; sz += z;
    delay(SAMPLE_DELAY_MS);
}

float wx = (float)sx / WINDOW_SIZE;
float wy = (float)sy / WINDOW_SIZE;
float wz = (float)sz / WINDOW_SIZE;

float dX = wx - rest_mean_x;
float dY = wy - rest_mean_y;
float dZ = wz - rest_mean_z;

int zone = classifyZone(dX);
String result;

/***** PERFECTED LOGIC *****/
if(!leftMode && !rightMode){ // STRAIGHT MODE
    if(zone == 0) result = "Front foot defense";
    else result = "bad shot";
}
else if(rightMode){ // RIGHT MODE
    bool lifted = (abs(dY) > Y_LIFT_MIN) || (abs(dZ) > Z_LIFT_MIN);
    if(zone == 2 && lifted) result = "Cover drive";
```

IoT-Driven Smart Bat for Real-Time Swing and Shot Evaluation

```
    else result = "bad shot";
  }
  else if(leftMode){ // LEFT MODE
    bool downward = (abs(dY) > Y_DROP_MIN) || (abs(dZ) > Z_DROP_MIN);
    if(zone == 1 && downward) result = "Left drive";
    else result = "bad shot";
  }

  // FEEDBACK HANDLING — only change feedback on transitions:
  if(result == "bad shot"){
    if(lastShotWasGood){
      // only when previous shot was good, show a NEW feedback
      showRandomFeedbackIfBad();
      lastShotWasGood = false;
    }
    // if lastShotWasGood == false, do nothing (keep existing feedback)
  } else {
    // correct shot: clear feedback and mark lastShotWasGood
    if(lastFeedbackIndex != -1){
      Blynk.virtualWrite(FEEDBACK_VPIN, ""); // clear feedback
      lastFeedbackIndex = -1;
    }
    lastShotWasGood = true;
  }

  send(result);
  lastTrigger = millis();
}

/***** showRandomFeedbackIfBad *****/
```

IoT-Driven Smart Bat for Real-Time Swing and Shot Evaluation

// Show a new random feedback only when called for a bad shot.

// Ensures it's different from the last shown feedback.

```
void showRandomFeedbackIfBad(){
    int newIdx = random(0, FEEDBACK_COUNT);
    // avoid repeating same feedback twice in a row
    if(FEEDBACK_COUNT > 1){
        int attempts = 0;
        while(newIdx == lastFeedbackIndex && attempts < 10){
            newIdx = random(0, FEEDBACK_COUNT);
            attempts++;
        }
    }
    lastFeedbackIndex = newIdx;
    Blynk.virtualWrite(FEEDBACK_VPIN, feedbacks[newIdx]);
    Serial.println(String("FEEDBACK: ") + feedbacks[newIdx]);
}
```

/****** checkMic helper *****/

```
void checkMic(int idx, int hwPin, int vPin){
    int state = digitalRead(hwPin);
    unsigned long now = millis();
    if(state != lastMicState[idx]){
        if(now - lastMicTime[idx] > MIC_DEBOUNCE_MS){
            lastMicTime[idx] = now;
            lastMicState[idx] = state;
            Blynk.virtualWrite(vPin, state ? 1 : 0);
            if(state == HIGH){
                String msg;
                if(idx==0) msg = "Top Edge tapped";
                else if(idx==1)
```

```
{
    msg = "Centre Shot tapped";
    // short buzzer pattern for centre
    digitalWrite(buzzer,HIGH);
    delay(50);
    digitalWrite(buzzer,LOW);
    delay(50);
    digitalWrite(buzzer,HIGH);
    delay(50);
    digitalWrite(buzzer,LOW);
    delay(50);
}
else msg = "Bottom Edge tapped";
Blynk.virtualWrite(MIC_V_STRING, msg);
Serial.println("MIC EVENT: " + msg);
}
}
}
}

/***** SEND *****/
void send(String s){
    Serial.println("RESULT: " + s);
    if(sendEnabled){
        Blynk.virtualWrite(RESULT_PIN, s);
    }
}

/***** CALIBRATION *****/
void calibrateRest(){
```

IoT-Driven Smart Bat for Real-Time Swing and Shot Evaluation

```
long sx=0, sy=0, sz=0;
for(int i=0;i<CAL_SAMPLES_REST;i++){
    sx += mpu.getRotationX();
    sy += mpu.getRotationY();
    sz += mpu.getRotationZ();
    delay(10);
}
rest_mean_x = (float)sx / CAL_SAMPLES_REST;
rest_mean_y = (float)sy / CAL_SAMPLES_REST;
rest_mean_z = (float)sz / CAL_SAMPLES_REST;

// reset smoothing buffers so averages align to the new rest baseline
sumX = sumY = sumZ = 0;
bufIndex = 0;
bufFilled = false;
for (int i=0;i<MOVING_BUF;i++){
    bufX[i]=bufY[i]=bufZ[i]=0;
}
// reset mic last states/time so we don't get a burst of events after recalibration
lastMicState[0] = digitalRead(MIC_TOP_PIN);
lastMicState[1] = digitalRead(MIC_CENTER_PIN);
lastMicState[2] = digitalRead(MIC_BOT_PIN);
unsigned long now = millis();
for(int i=0;i<3;i++) lastMicTime[i] = now;

Serial.println("Calibration complete. rest_mean_x=" + String(rest_mean_x) +
    " rest_mean_y=" + String(rest_mean_y) + " rest_mean_z=" + String(rest_mean_z));
}

/***** BUFFER SMOOTHING *****/
```

IoT-Driven Smart Bat for Real-Time Swing and Shot Evaluation

```
void updateBuffer(int16_t x,int16_t y,int16_t z){
    sumX -= bufX[bufIndex];
    sumY -= bufY[bufIndex];
    sumZ -= bufZ[bufIndex];

    bufX[bufIndex] = x;
    bufY[bufIndex] = y;
    bufZ[bufIndex] = z;

    sumX += x;
    sumY += y;
    sumZ += z;

    bufIndex++;
    if(bufIndex >= MOVING_BUF){
        bufIndex = 0;
        bufFilled = true;
    }
}

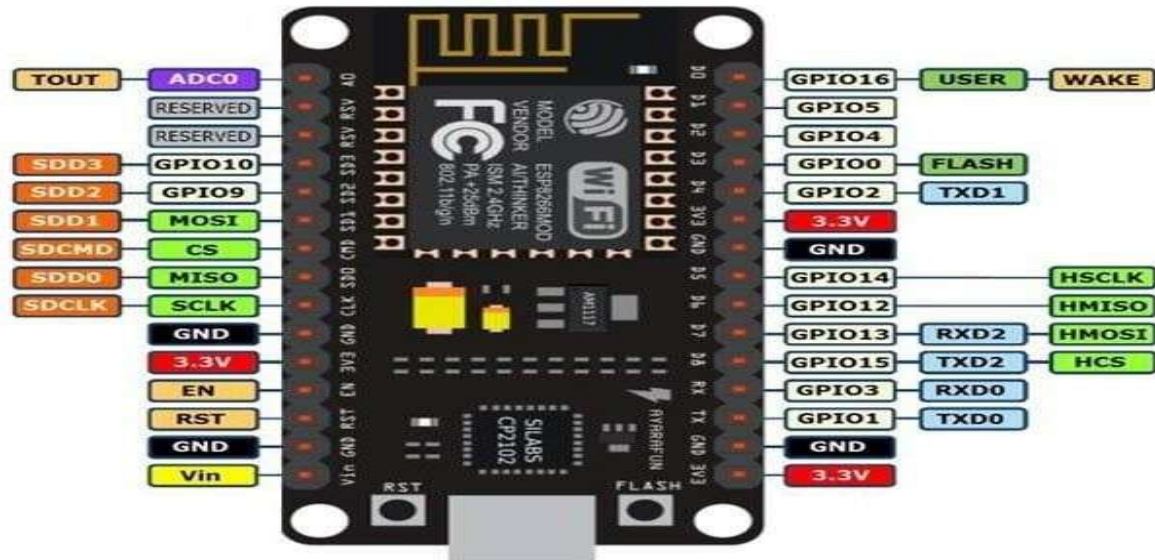
float getAvgX(){
    int denom = bufFilled ? MOVING_BUF : (bufIndex==0 ? 1 : bufIndex);
    return (float)sumX / denom;
}

float getAvgY(){
    int denom = bufFilled ? MOVING_BUF : (bufIndex==0 ? 1 : bufIndex);
    return (float)sumY / denom;
}

float getAvgZ(){
    int denom = bufFilled ? MOVING_BUF : (bufIndex==0 ? 1 : bufIndex);
    return (float)sumZ / denom;}
```

APPENDIX C

SPECIFICATION OF NodeMCU



❖ The General Specifications

- 1) Microcontroller: ESP8266EX
- 2) Module Type: NodeMCU Development Board
- 3) Architecture: 32-bit
- 4) CPU Frequency: 80 MHz
- 5) Flash Memory: 4 MB
- 6) Operating System Support: RTOS / Non-OS

❖ Electrical Specifications

- 1) Operating Voltage: 3.3 V
- 2) Input Voltage (VIN): 4.5 V – 10 V
- 3) GPIO Voltage Level: 3.3 V
- 4) Power Consumption: Low power with sleep modes

❖ **Communication Interfaces**

- 1) UART
- 2) SPI
- 3) I²C

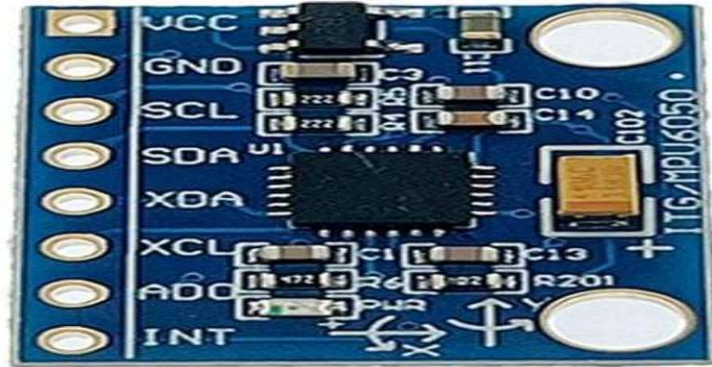
❖ **Pin Configuration**

- 1) Digital GPIO Pins: Multiple
- 2) Analog Input Pin: A0
- 3) Power Pins: VIN, 3V3, GND
- 4) Communication Pins: TX, RX, SDA, SCL

❖ **Features**

- 1) Built-in Wi-Fi (IEEE 802.11 b/g/n)
- 2) TCP/IP protocol stack
- 3) Supports IoT cloud platforms

SPECIFICATION OF MPU6050



❖ General Specifications

- 1) Sensor Type: Motion tracking device
- 2) Axes: 6-axis (3-axis accelerometer + 3-axis gyroscope)
- 3) Manufacturer: InvenSense
- 4) Package Type: Module
- 5) Built-in Feature: Digital Motion Processor (DMP)

❖ Electrical Specifications

- 1) Operating Voltage: 3.3 V
- 2) Interface Voltage: 3.3 V
- 3) Power Consumption: Low power

❖ Pin Configuration

- 1) VCC: Power supply
- 2) GND: Ground
- 3) SDA: Serial Data Line
- 4) SCL: Serial Clock Line
- 5) INT: Interrupt output (optional)

SPECIFICATION OF MICROPHONE



❖ General Specifications

- 1) Sensor Type: Electret microphone module
- 2) Function: Sound and vibration detection

- 3) Output Type: Analog
- 4) Sensitivity: High sensitivity

❖ **Electrical Specifications**

- 1) Operating Voltage: 3.3 V – 5 V
- 2) Output Voltage: Analog signal proportional to vibration
- 3) Current Consumption: Low
- 4) Frequency Characteristics
- 5) Frequency Response: Wide frequency range
- 6) Adjustable Gain: Yes (on-board potentiometer)

❖ **Pin Configuration**

- 1) VCC: Power supply
- 2) GND: Ground
- 3) OUT: Analog output

SPECIFICATION OF BUZZER



Buzzer

A **buzzer module** is an audio signaling device that produces sound when energized. It is commonly used in embedded systems for alerts, alarms, status indicators, and feedback tones.

1. General Features

- **Type:** Piezoelectric Buzzer / Electromechanical Buzzer (depends on model)
- **Function:** Converts electrical signals into audible sound
- **Module Type:** Active or Passive
 - **Active Buzzer:** Contains built-in oscillator; emits sound with DC signal
 - **Passive Buzzer:** Requires external PWM or tone signal to generate sound

2. Electrical Characteristics

- **Operating Voltage:** 3.3V to 5V DC (standard modules)
- **Current Consumption:**
 - Active: ~10–30 mA
 - Passive: ~5–15 mA
- **Power Rating:** < 0.5W

3. Sound Characteristics

- **Sound Output:** 85 dB to 100 dB @ 10 cm (depends on model)
- **Resonant Frequency:**
 - Active: Typically fixed around 2 kHz to 4 kHz
 - Passive: Frequency-controlled by the user (typically 500 Hz – 5 kHz)
- **Tone Type:** Continuous or modulated (user-controlled in passive buzzer)

4. Pin Configuration

- **VCC/+:** Positive power supply (3.3V–5V)
- **GND/-:** Ground
- **I/O Signal (sometimes shared with VCC in active buzzers)**

5. Environmental Conditions

- **Operating Temperature:** -20°C to +70°C
- **Storage Temperature:** -30°C to +80°C

6. Applications

- Alarm systems
- Timers and reminders
- Notification systems
- Keypad feedback
- Line-following and obstacle-avoidance robots

BATTERY



Rechargeable Battery

Rechargeable batteries are used to provide portable and reusable power to electronic circuits. One of the most common types used in embedded projects is the **Lithium-ion (Li-ion)** or **Lithium Polymer (Li-Po)** battery due to its high energy density and rechargeability.

1. General Features

- **Battery Type:** Lithium-ion (Li-ion) / Lithium-Polymer (Li-Po)
- **Rechargeable:** Yes
- **Form Factor:** Cylindrical (e.g., 18650) or flat pouch (Li-Po)

2. Electrical Specifications

- **Nominal Voltage:** 3.7V (single cell)
- **Full Charge Voltage:** 4.2V
- **Cut-off Voltage (Discharge Limit):** ~3.0V
- **Capacity:** Typically ranges from 1000 mAh to 3000 mAh (e.g., 18650 = 2200 mAh)
- **Discharge Current:**
 - **Standard:** 0.5C to 1C (e.g., 1C = 2200 mA for 2200 mAh battery)
 - **Maximum:** Up to 2C to 3C for some high-drain models

3. Charging Specifications

- **Recommended Charging Voltage:** $4.2V \pm 0.05V$
- **Charging Current:** ~0.5C to 1C
- **Charging Method:** Constant Current / Constant Voltage (CC/CV)



HKBK COLLEGE OF ENGINEERING, Bengaluru - 45
Department of Electronics and Communication
Engineering
Project Budget

Project Title: Smart Medicine Consumption Tracker for Patients Using AI and IOT

Team Members: Kiran kumar goudar, melwin jain, Mutturaj, and Nithin S.S

Guide Name: Shruthi B M

Group Number: 28

S. No.	Item Description	Quantity	Cost per Unit (₹)	Total Cost (₹)
1	NodeMCU	1	1000	1000
2	MPU6050	10	1000	1000
3	microphone	3	250	750
4	Buzzer	1	50	50
5	Bat	1	1000	1000
6	Jumper Wires and Breadboard	1 set	450	450
7	Power Supply (Battery/Power Bank)	1	1000	1000
8	Enclosure and Miscellaneous	1	1000	1000
		Total Amount (₹)		6250

Guide Signature